

Different brains for different songs

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THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

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Ecology & Evolutionary Biology

Career goal:
Botanist

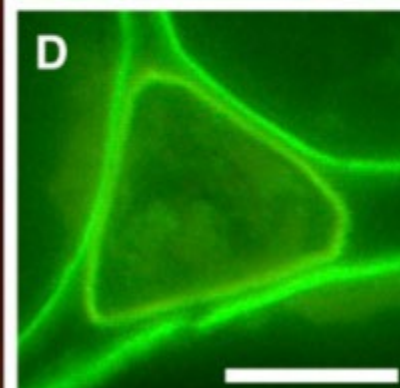
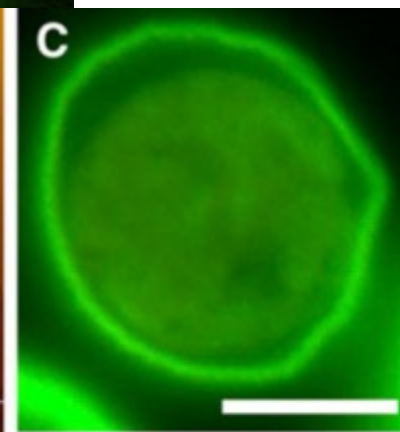
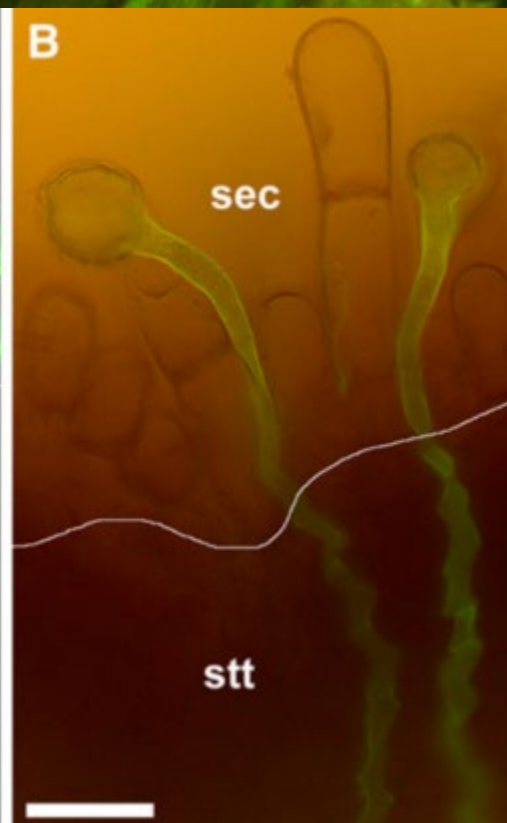
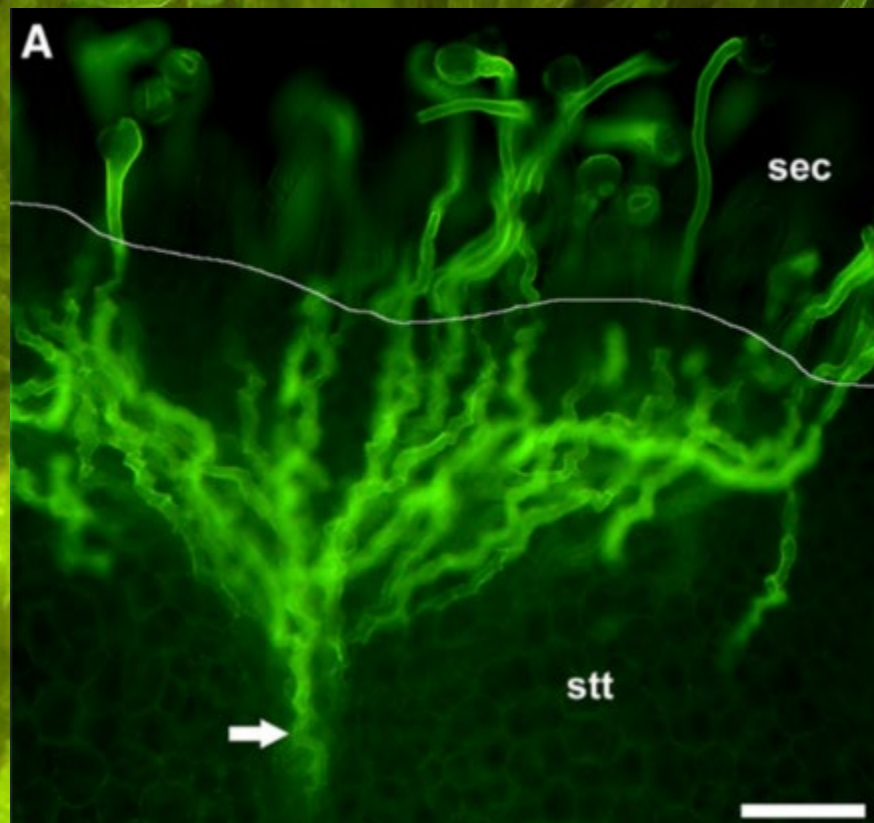
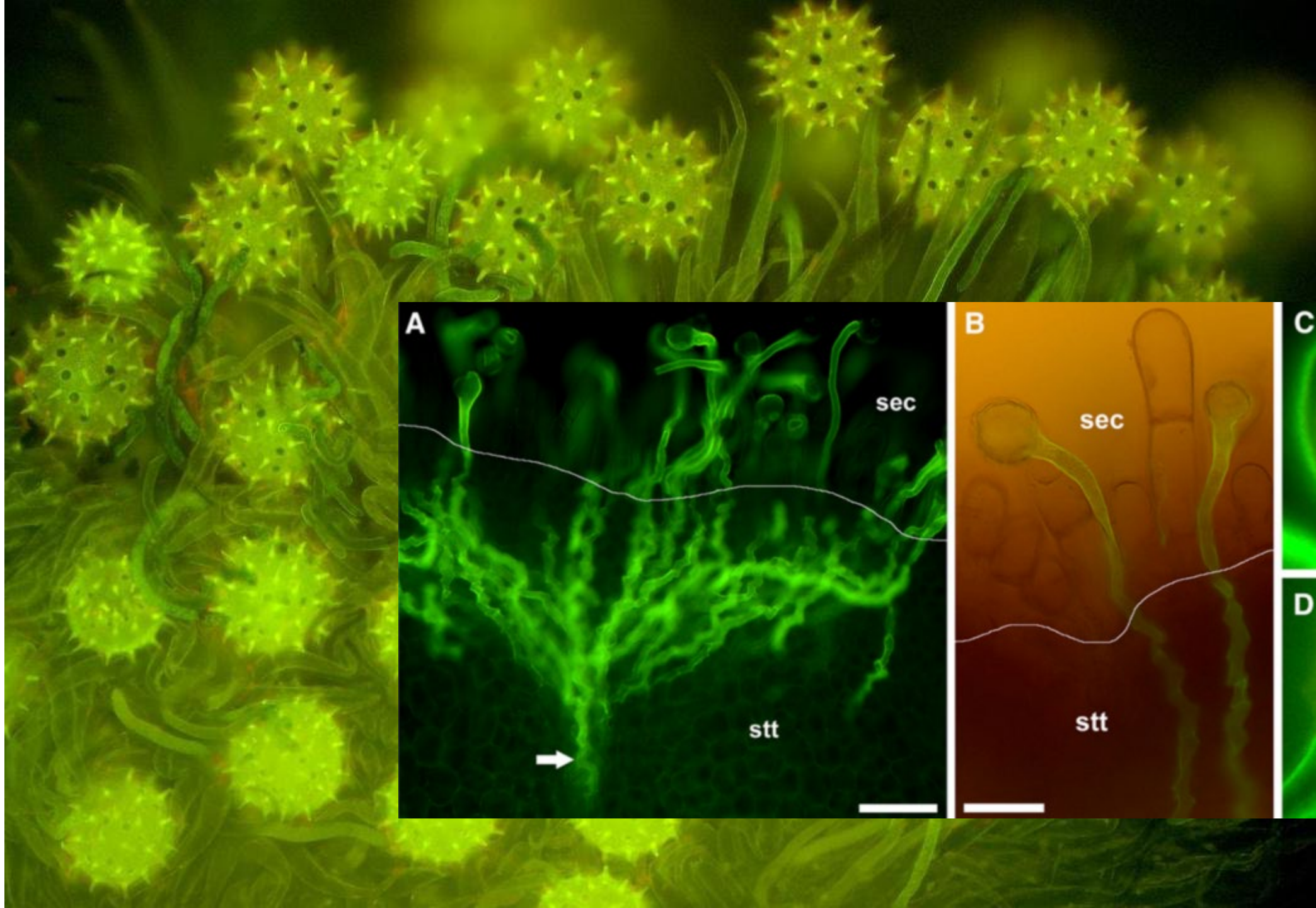
Spoiler alert:
Didn't happen

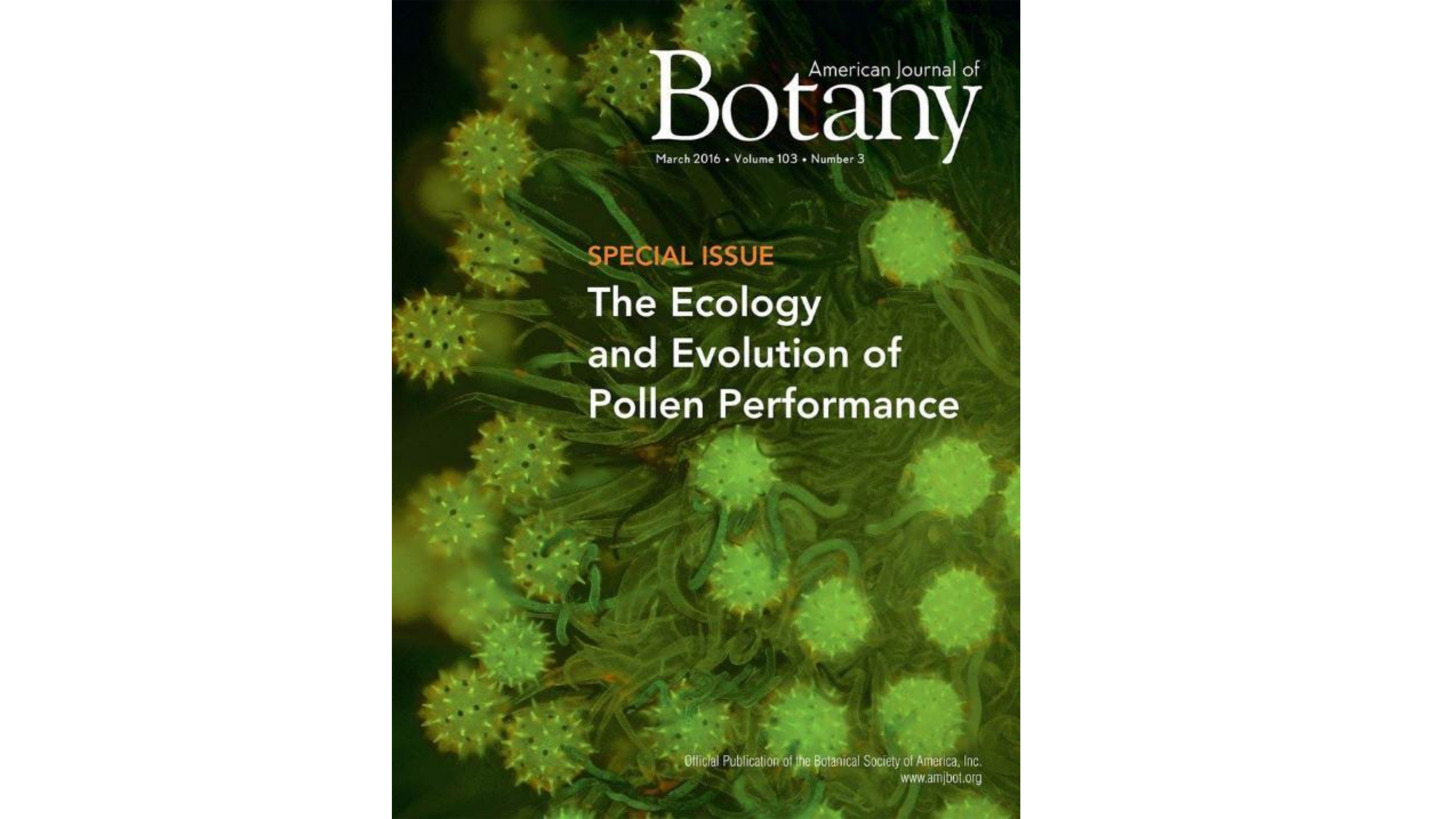


Best skills I learned:

- **Taking detailed notes**
- **Thinking methodically**
- **Staining tissue/using a microscope**





The background of the cover is a high-magnification micrograph of numerous green, spherical pollen grains. Each grain is covered in fine, radiating spines or apertures, giving them a spiky appearance. They are clustered together, with some in sharp focus and others blurred in the background.

American Journal of Botany

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SPECIAL ISSUE

The Ecology and Evolution of Pollen Performance

Official Publication of the Botanical Society of America, Inc.
www.amjbot.org



FOC | Family List | FOC Vol. 17 | Lamiaceae * | Coleus *

3. *Coleus bracteatus* Dunn, Notes Roy. Bot. Gard. Edinburgh. 8: 158. 1913.

光萼鞘蕊花 guang e qiao rui hua

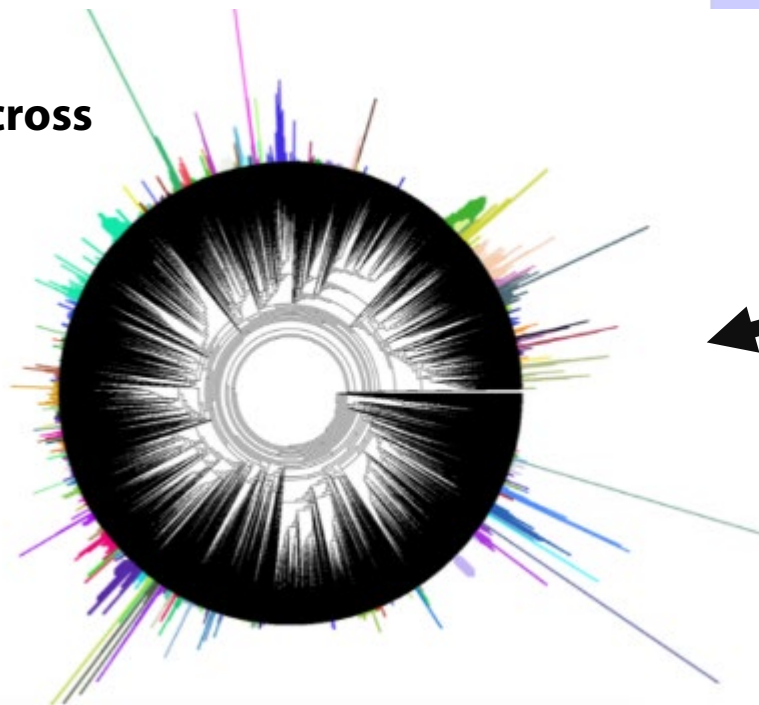
Herbs. Stems erect, to 90 cm, $\pm$ succulent, puberulent or subglabrous. Petiole slender, 2.5-5 cm; leaf blade ovate, 4.5-14.5 $\times$ 3-7 cm, membranous, puberulent or subglabrous, base broadly cuneate to shallowly cordate, margin crenate, apex acuminate. Panicles lax, terminal, to 42 cm, pubescent, verticillasters many flowered; peduncle 1-3 mm; bracts early deciduous. Pedicel 4-6 mm. Calyx campanulate, 3-4 mm in flower, to 8.5 mm in fruit, subglabrous or puberulent, densely rust colored glandular; teeth subequal, posterior 1 ovate, apex acute, reflexed in fruit; lateral and anterior teeth lanceolate, apex acuminate, straight. Corolla purple to reddish, puberulent, rust colored glandular, 1.6-1.8 cm; tube much exerted, abruptly reflexed beyond calyx, throat to 2 mm wide; limb 1/8-1/6 as long as tube, upper lip erect; lower lip slightly elongated, concave. Nutlets ovoid-globose. Fl. Sep.

* Forest margins, open areas, fields, rocky areas, wet forests; 1000-2200 m. Yunnan.

Flora of China
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Volume
Family
Genera
Advanced
SearchFlora of China
Home

Checklist

Flower size across
4,828 species



```

1  #!/usr/bin/env python
2
3  # will travel to a website and return a list of all links on the page
4
5  from bs4 import BeautifulSoup
6  import urllib, re
7
8  # open a file for output and accept a page to spyder
9  pageName = 'http://efloras.org/'
10 pageFileName = re.sub('http://', '', pageName)
11 pageFileName = re.sub('.+[/].+', '', pageName)
12 sourceFileName = 'url_list_%s.txt' % pageFileName
13 sourceFile = open(sourceFileName, 'w')
14
15 print 'Beginning to crawl ' + pageName
16
17 # put it in the soup
18 pageSource = BeautifulSoup(urllib.urlopen(pageName))
19
20 linksToTry = {}
21
22 for i in linksToTry.keys():
23     try:
24         # open a new page here
25         newPageSource = BeautifulSoup(urllib.urlopen(i))
26         print 'Attempting to check : ' + i
27         # get its links and add them to the list of linksToTry
28         for j in newPageSource.find_all('a'):
29             j = str(j.get('href'))
30             if not j.startswith('http://'):
31                 j = pageName + j
32             if j.startswith(pageName) and not j.startswith(pageName + '.java'):
33                 linksToTry[j] = ''
34             # add the tried link to running total
35             sourceFile.write(i+'\n')
36         except:
37             print 'Error attempting to open ' + i
38
39 # fin!
40 print 'All done! Open "homesrc.txt" to view results'
41
42 sourceFile.close()

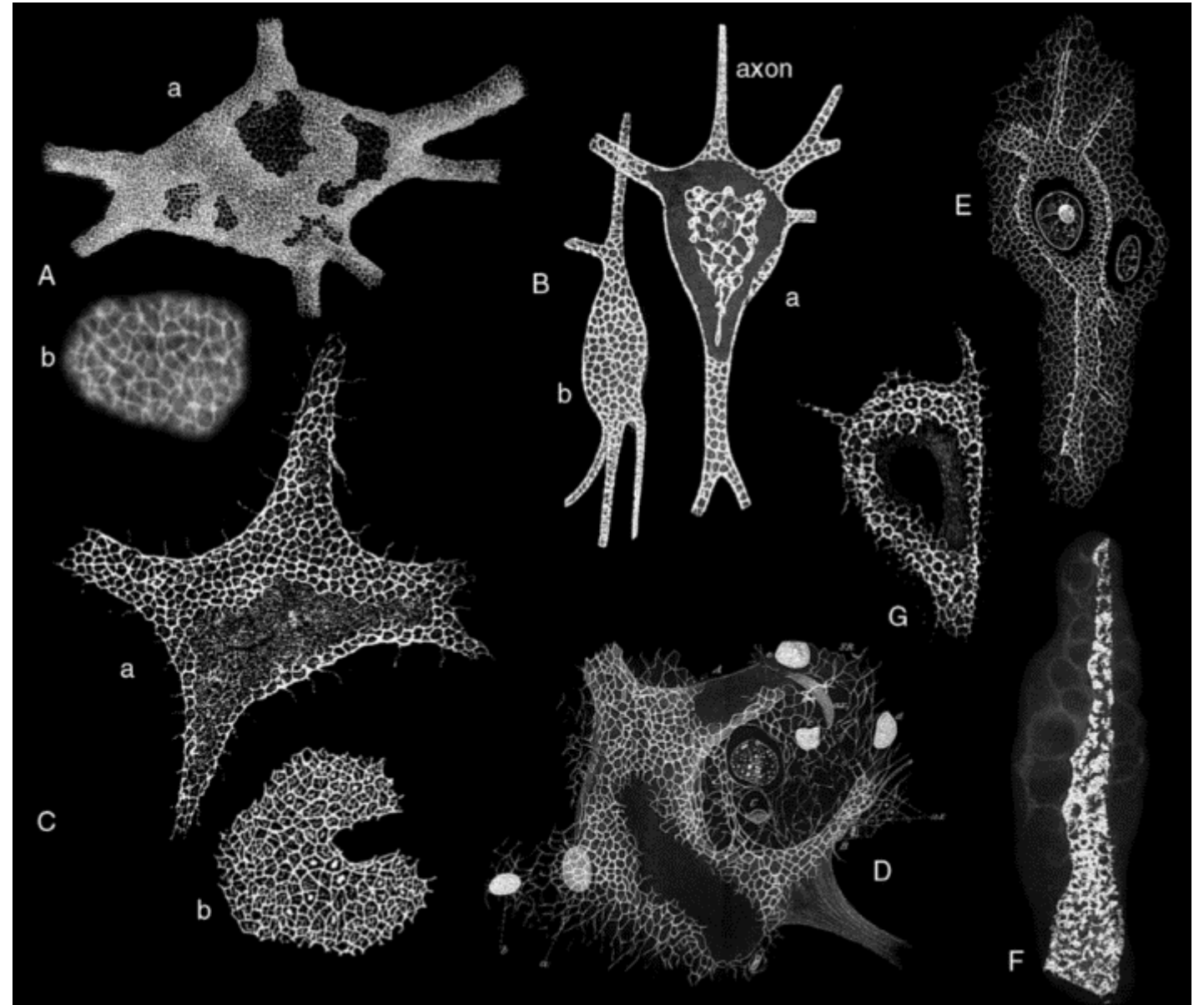
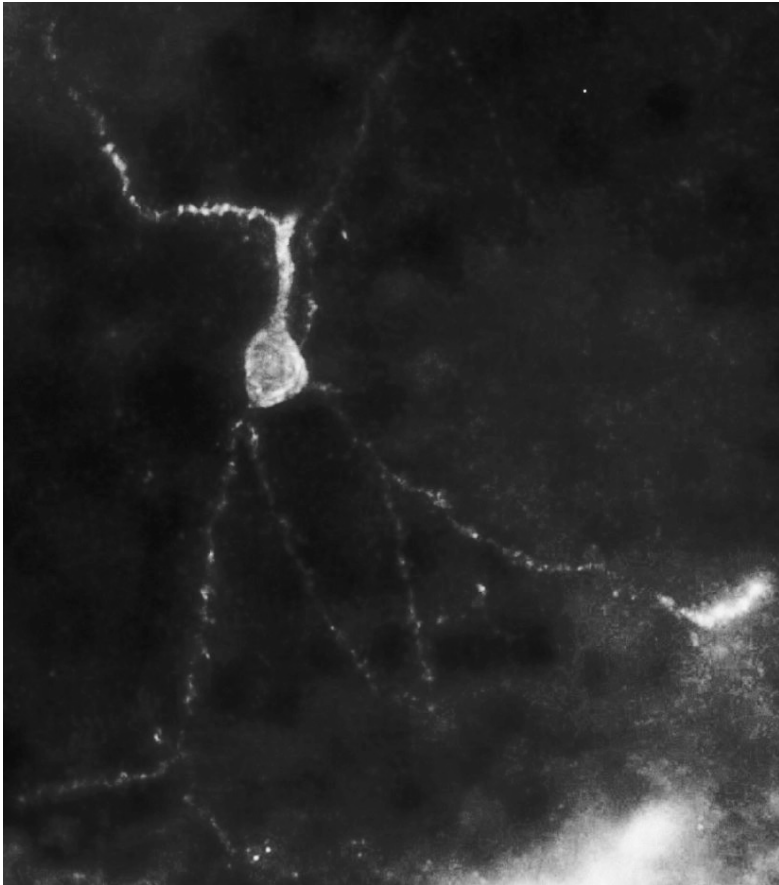
```

Searched
116,895 species

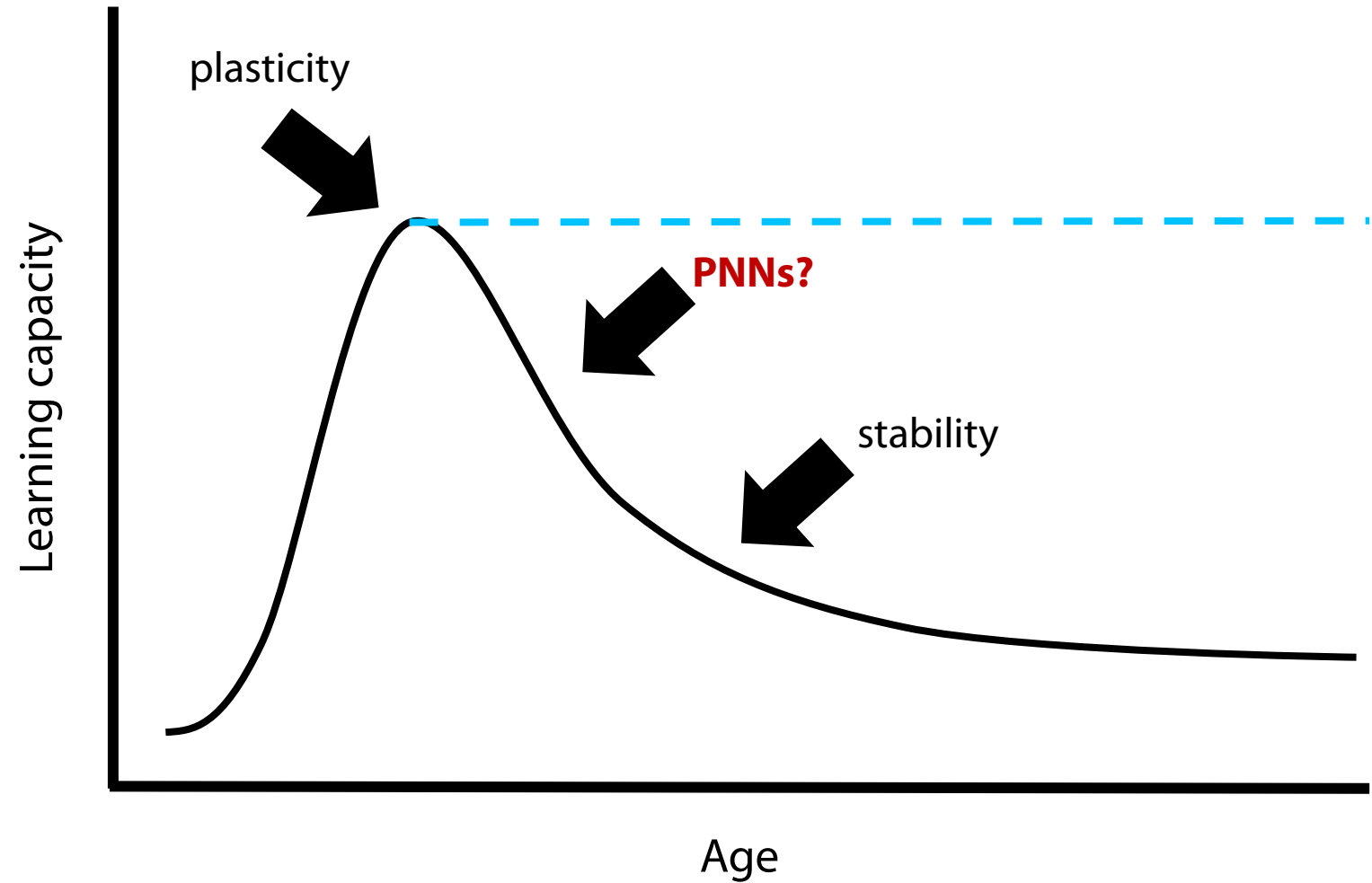
Perineuronal nets



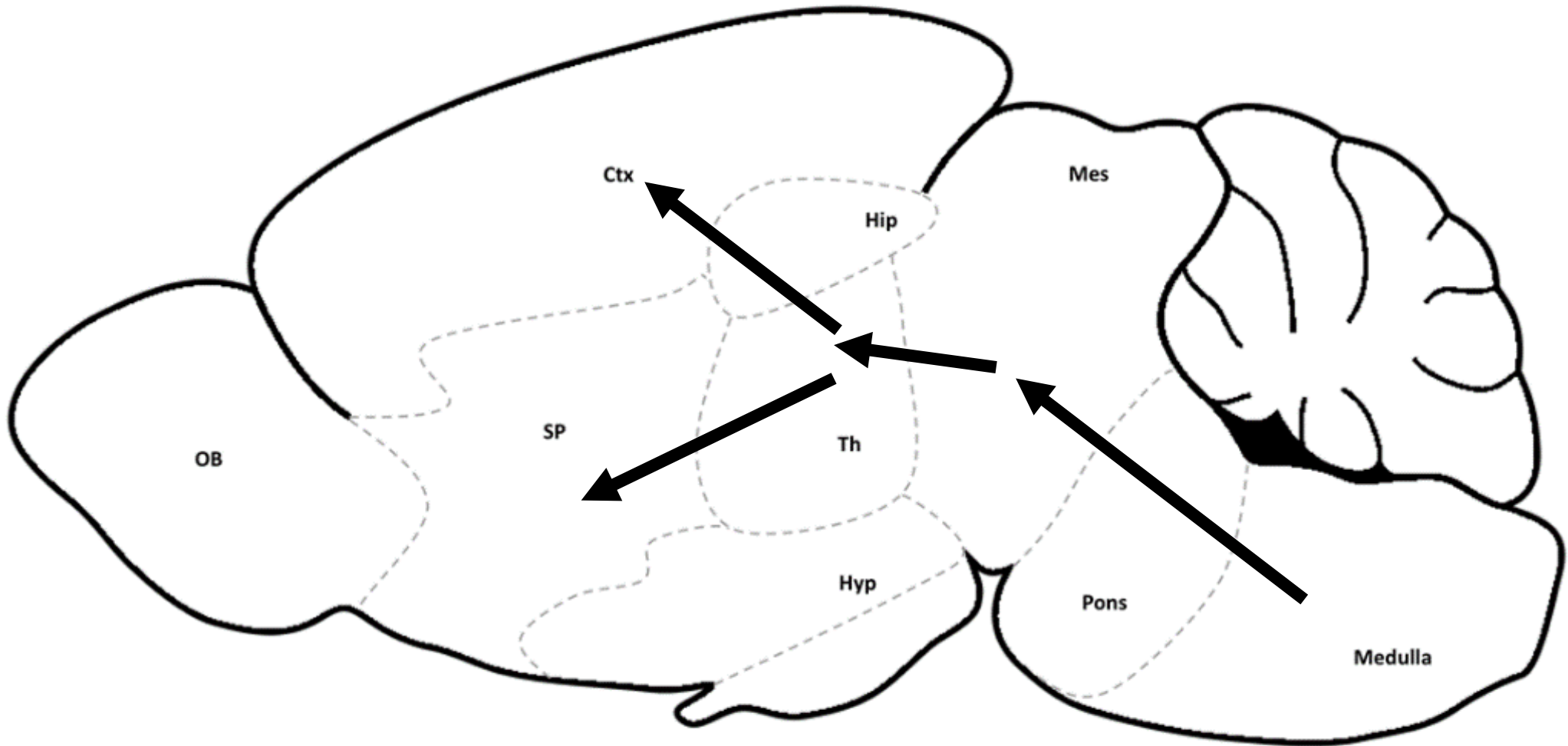
Colorado State University



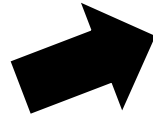
Learning gets harder as we get older



PNNs develop in the brain from the “bottom up”



Stability supports plasticity





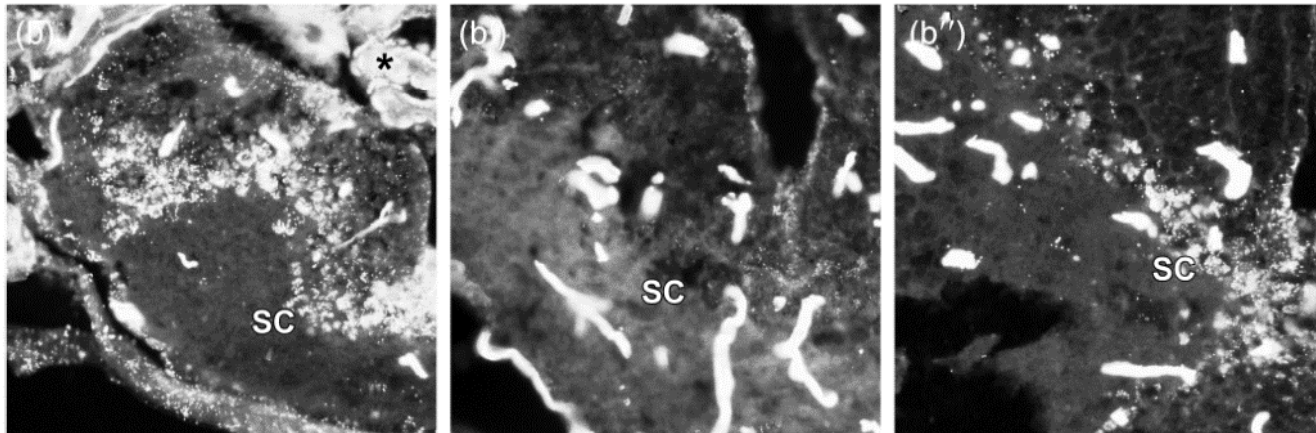
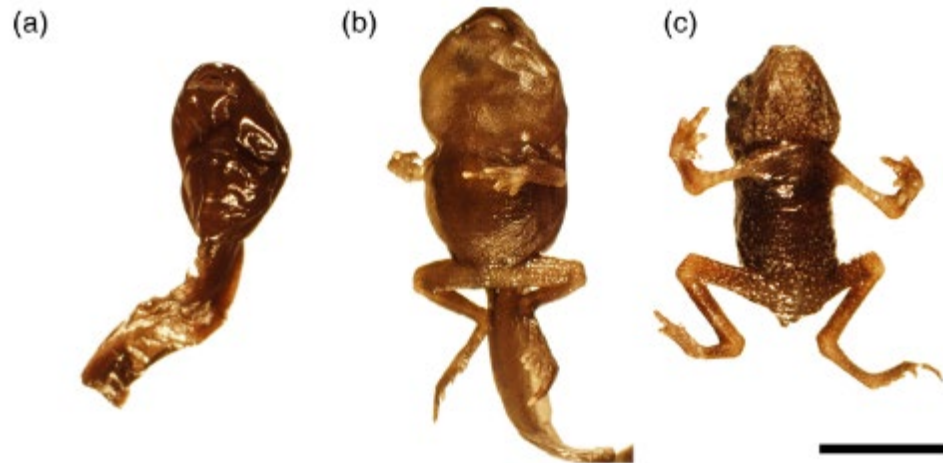
RESEARCH ARTICLE



WILEY

Dynamics of perineuronal nets over amphibian metamorphosis

Jacob A. Edwards | Makayla Risch | Kim L. Hoke



PNNs in the auditory system remained mysterious

Tactile

Visual

Auditory

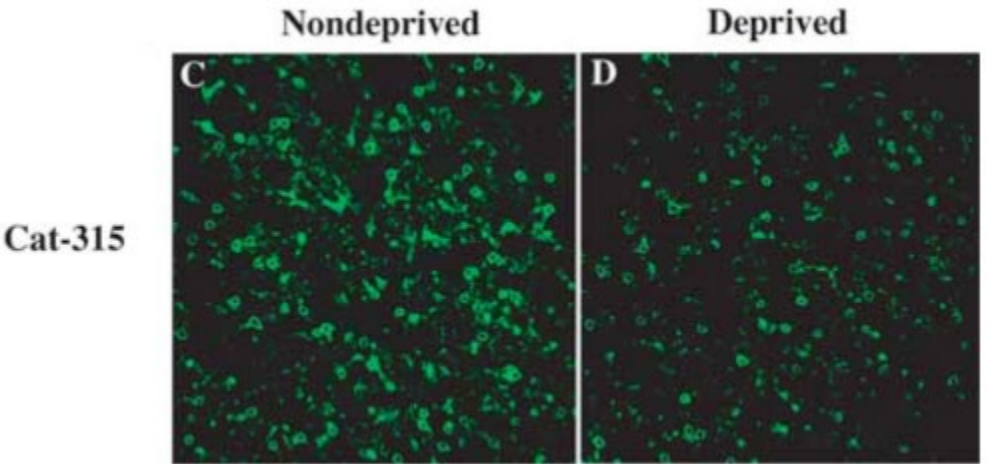
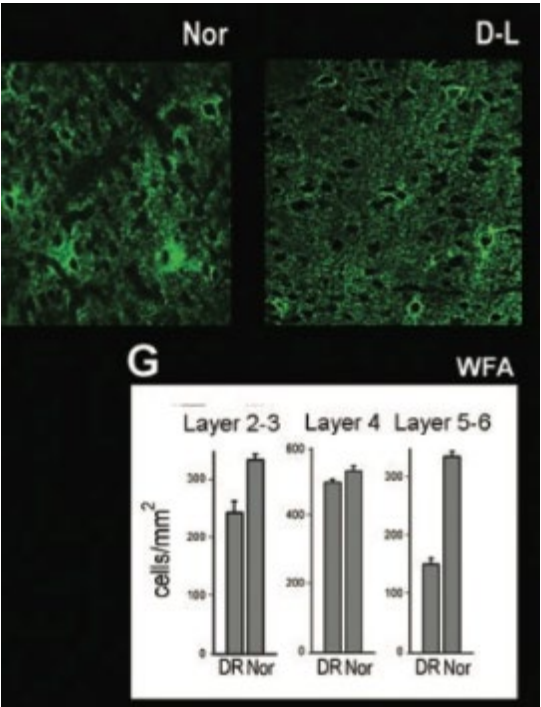


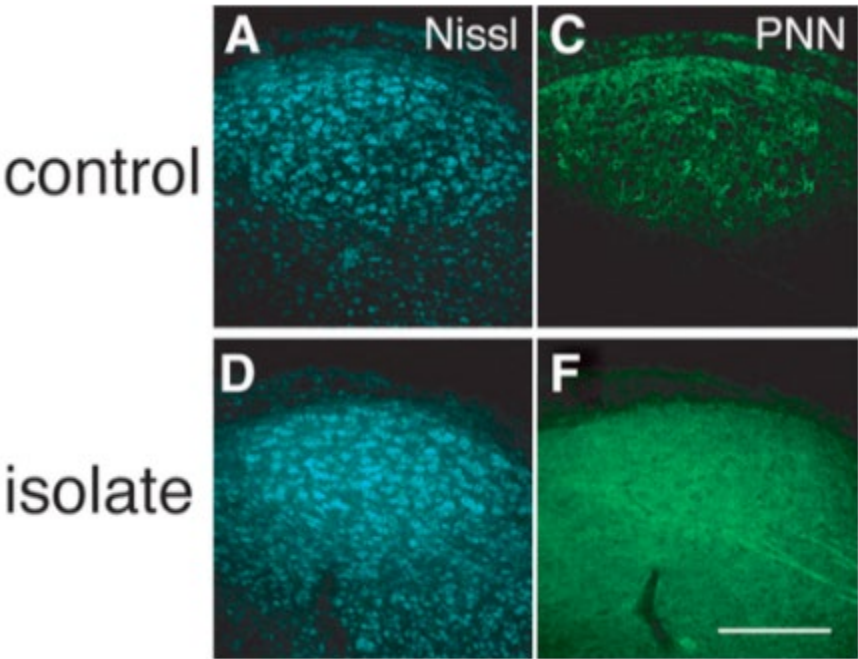
Table 1. Stereological cell counts of percentage of Cat-315 positive to total cell number

Group	n	Nondeprived (right), percentage of cells that are Cat-315+ ($\pm$ SD)	Deprived (left), percentage of cells that are Cat-315+ ($\pm$ SD)	p value (right vs left)
P30 control	11	7.419125 $\pm$ 1.851511	7.589479 $\pm$ 1.462587	0.3878
P30 Trim	10	7.395954 $\pm$ 1.78214	5.769441 $\pm$ 1.304703*	0.0034
Adult control	4	5.598945 $\pm$ 0.682995**	5.67208 $\pm$ 0.940417**	0.8708
Adult Trim	5	6.845911 $\pm$ 0.697807	6.768525 $\pm$ 0.794268	0.8673

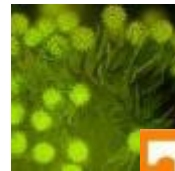
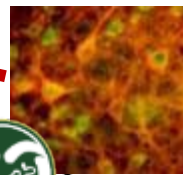
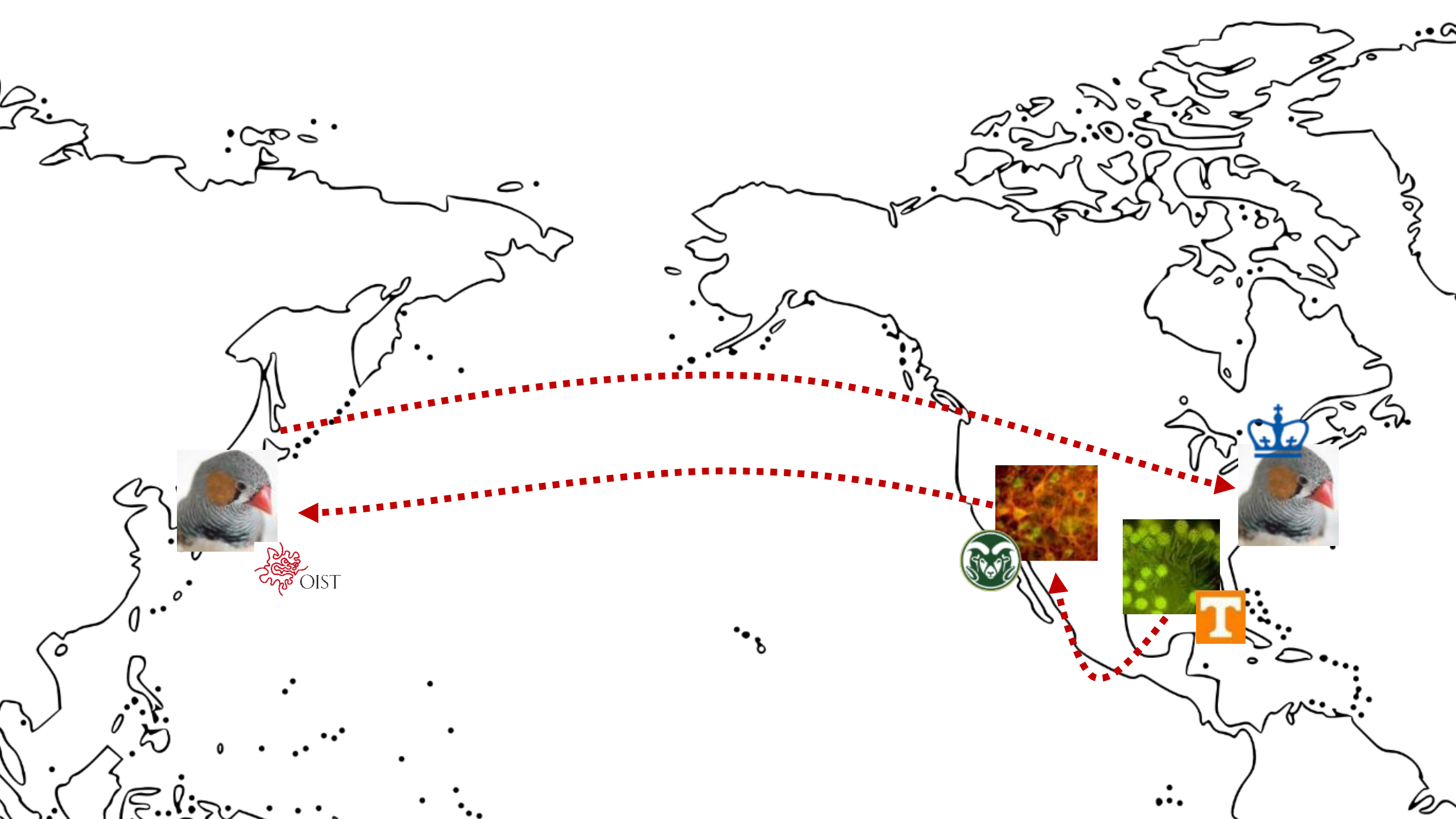
McRae *et al.* 2007



Pizzorusso *et al.* 2002



Balmer *et al.* 2009



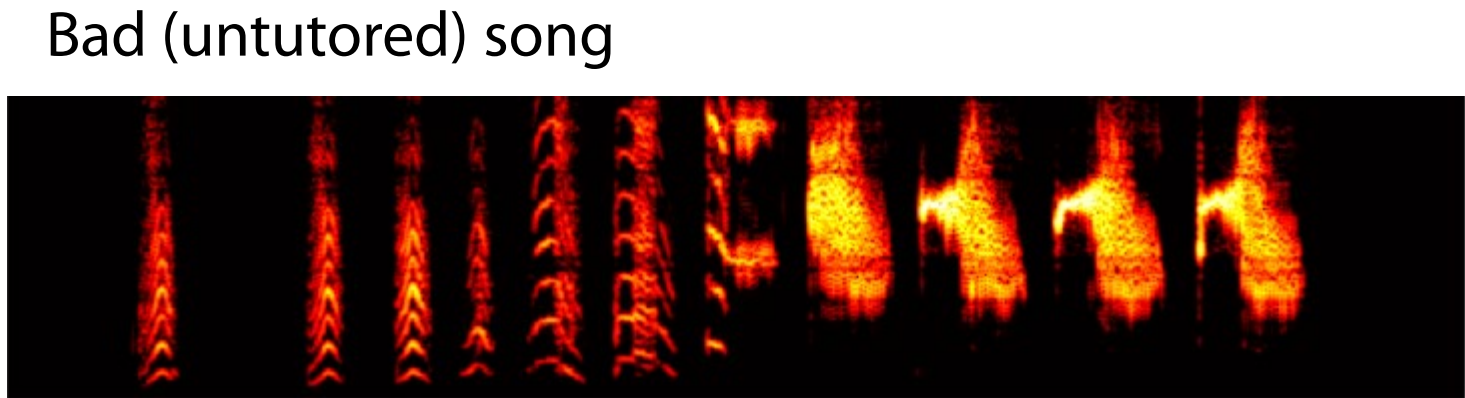
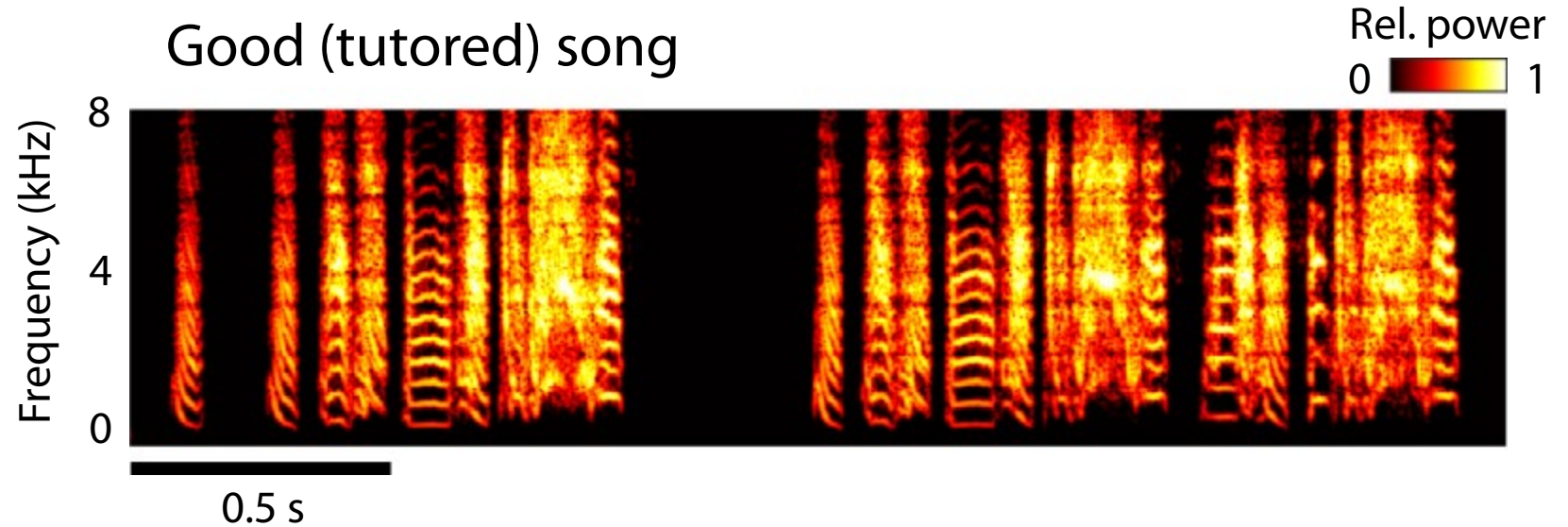
Jacob's life advice (your mileage may vary):

- **Talk to people!**
- **Dive deep into your interests;** follow your curiosity as your interests evolve
- **Learn how to write code.** Python is a great 1st language (it helps to pick a project that is meaningful to you). Writing code forces you to think methodically

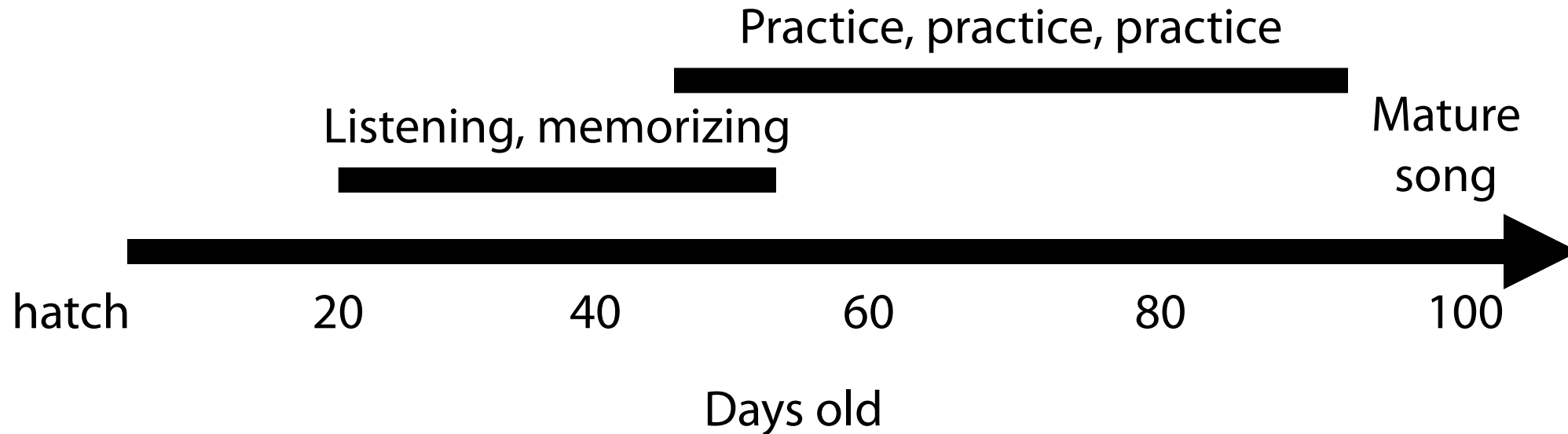
The power of listening



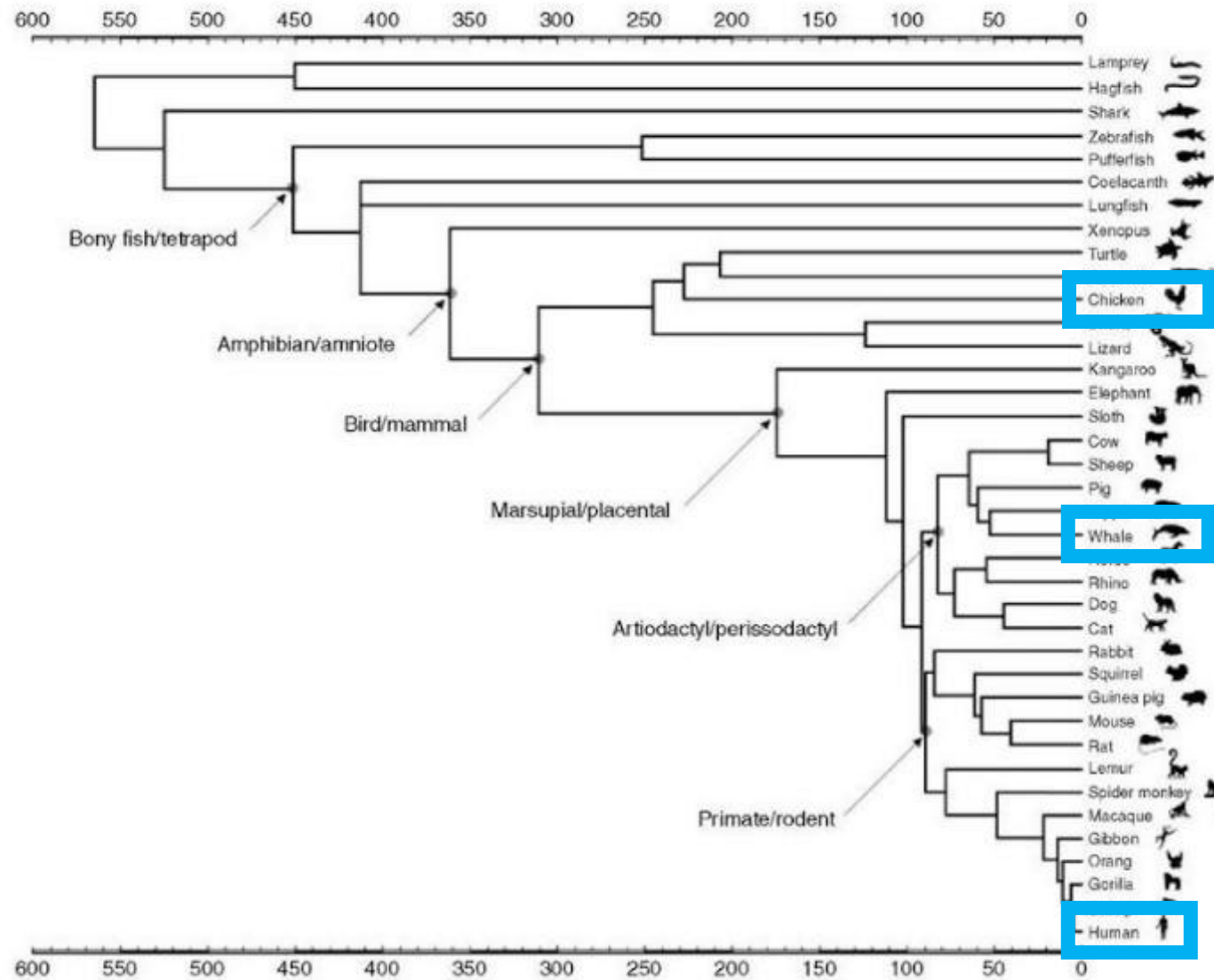
Song is essential for finding a mate



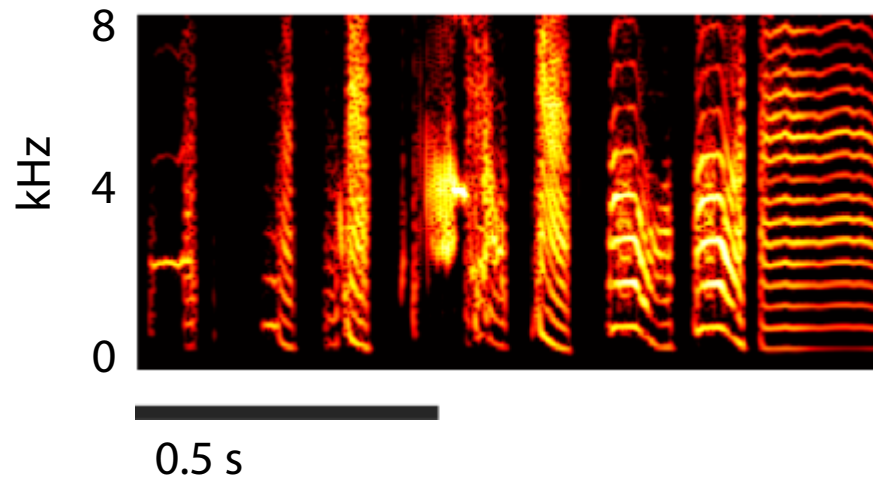
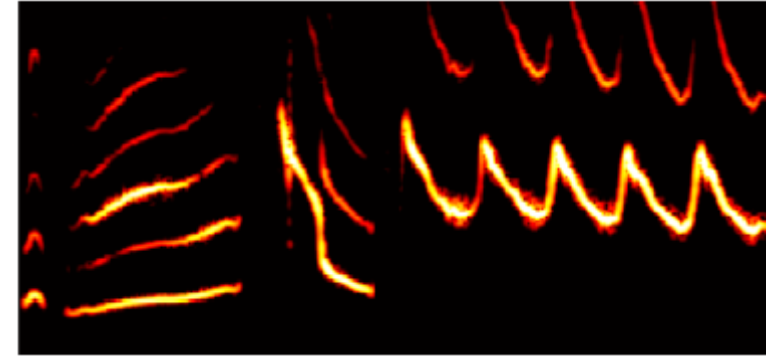
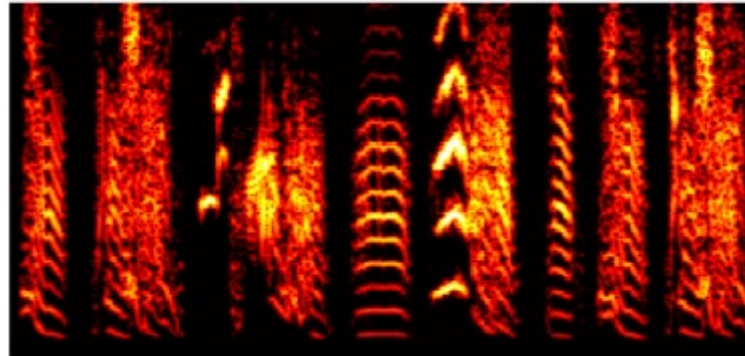
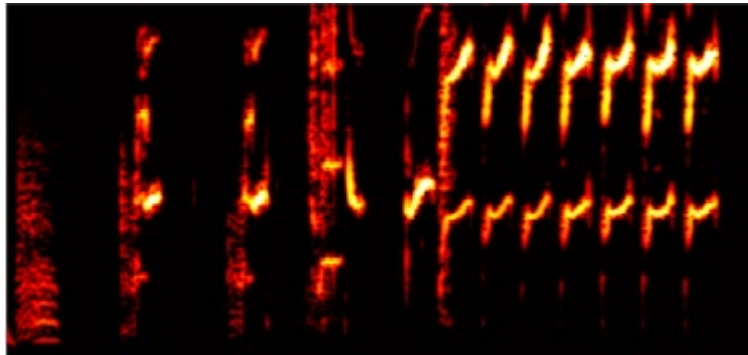
Songbirds are outstanding vocal learners



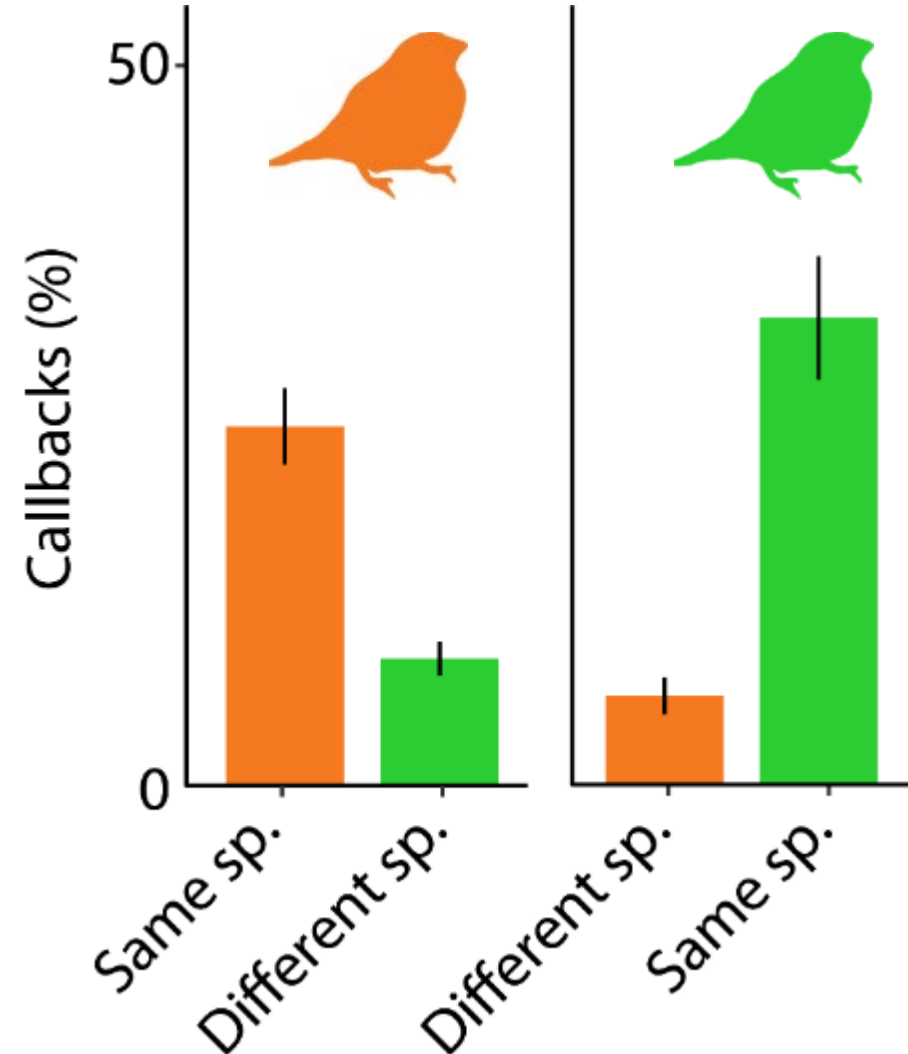
Vocal learning is RARE



Song is individual-specific and species-specific

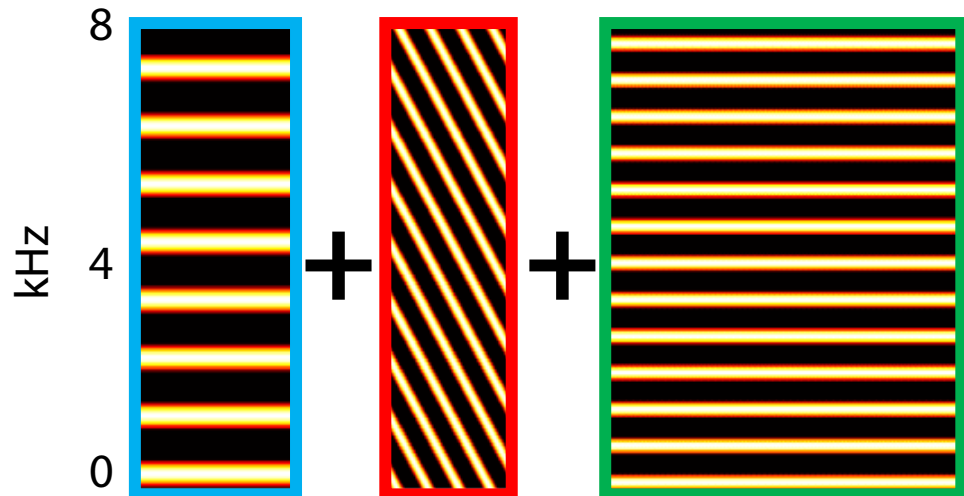
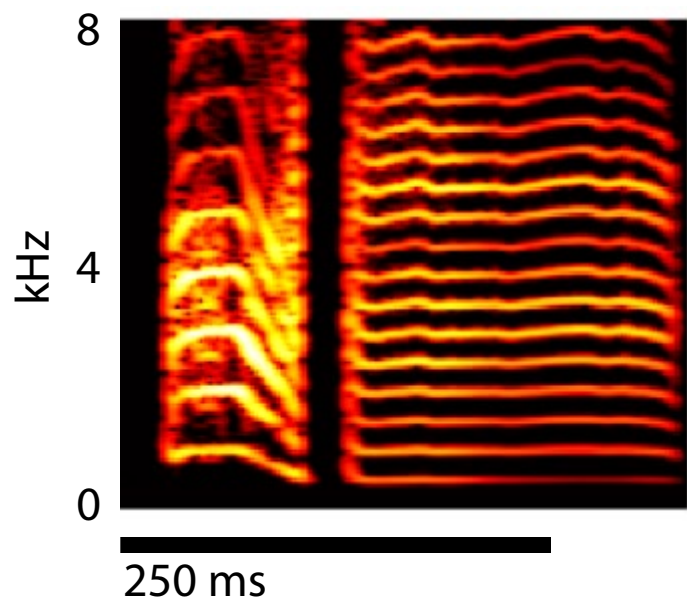


In behavior, songbirds are most interested in their own species (conspecifics)

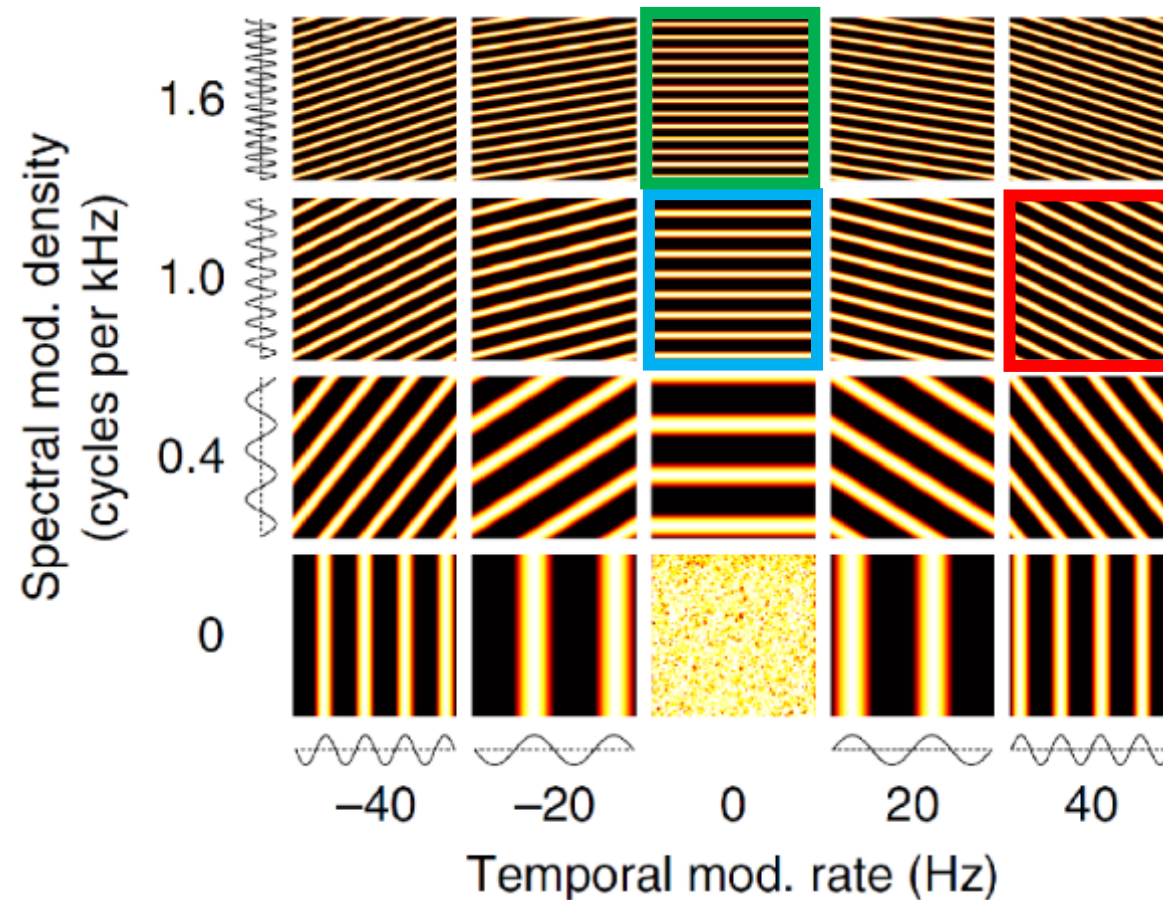


Data from labmate Moises Rivera

Describing the acoustic structure of song



1. **Density** of harmonics
2. **Change** in freq. **over time**



Hypothesis: Birds key in on acoustic structure of conspecifics

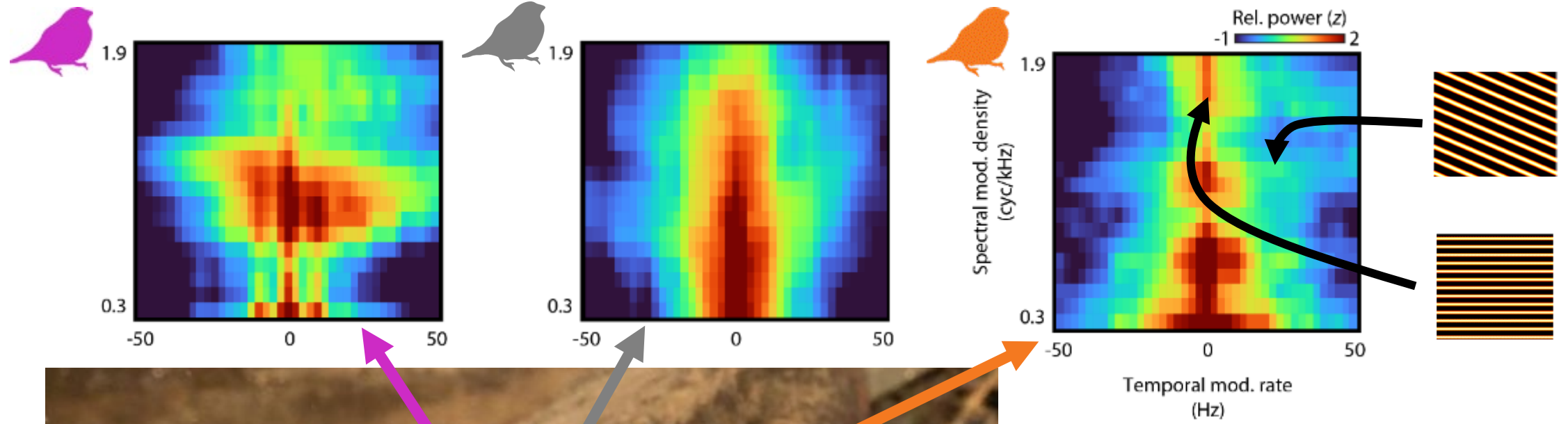
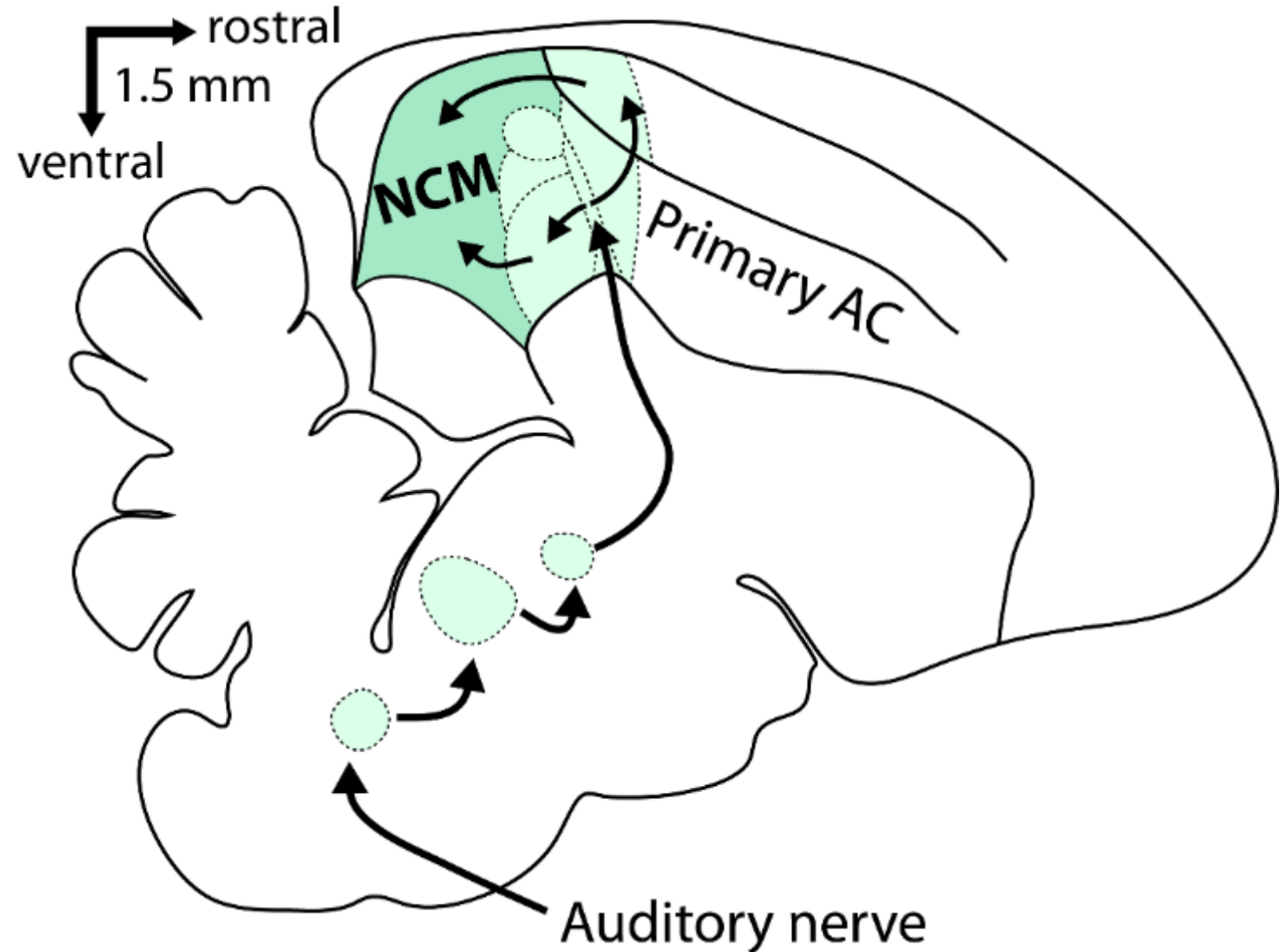
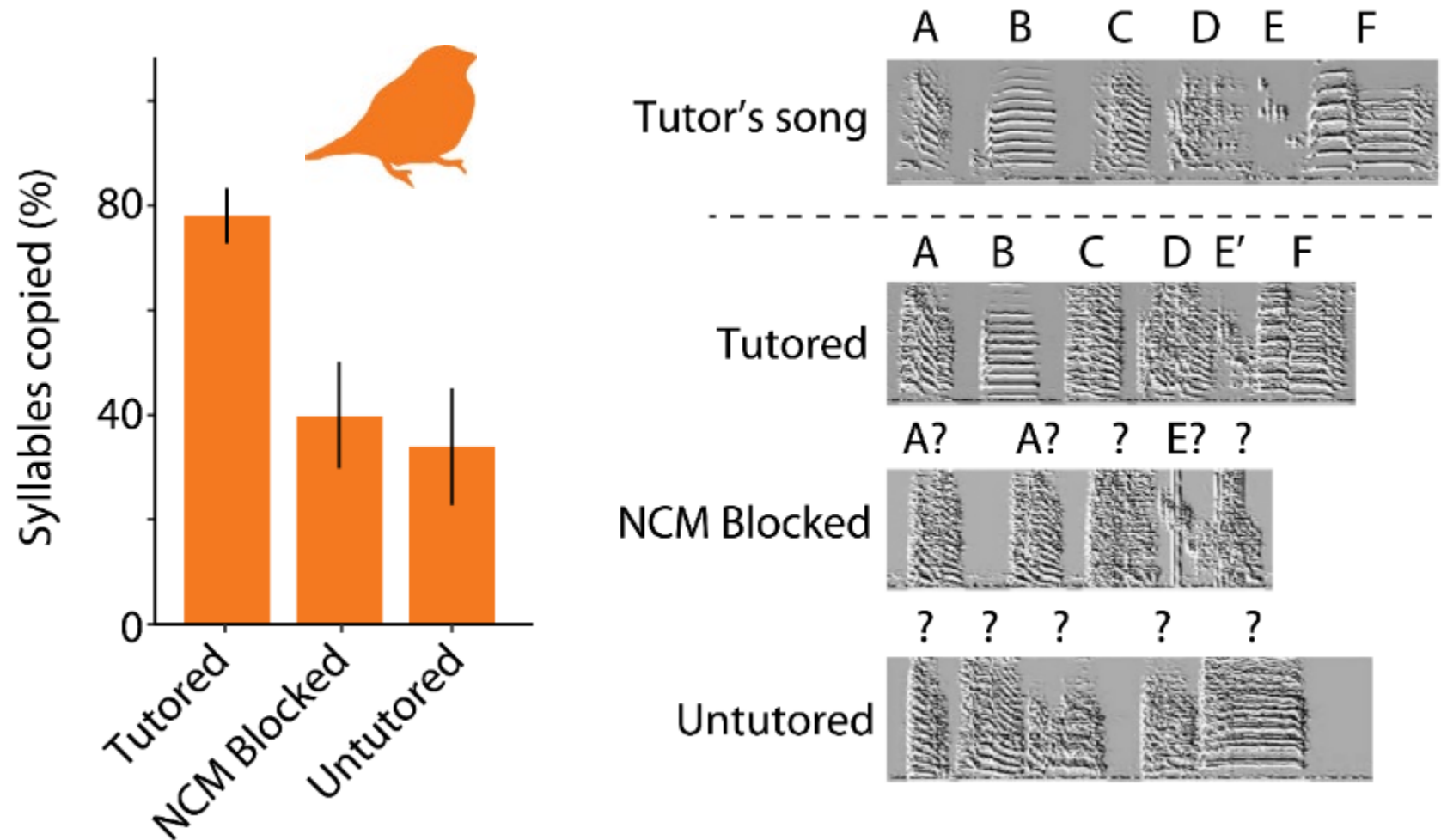


Photo credit:
Col Roberts Photography

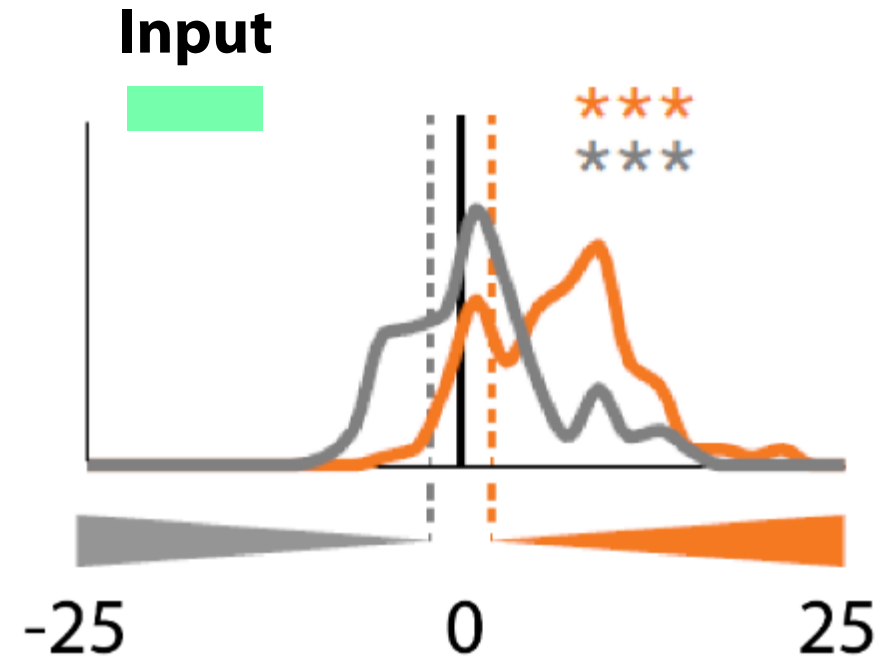
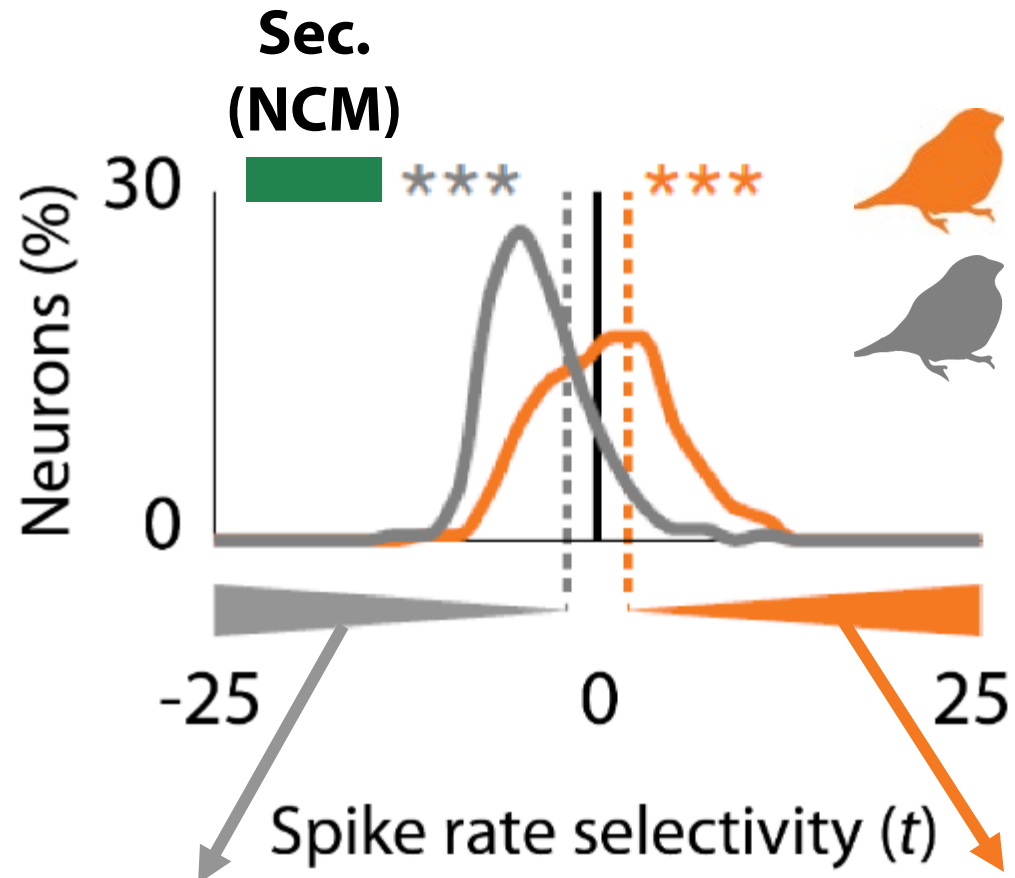
Where are songs encoded for conspecific communication?



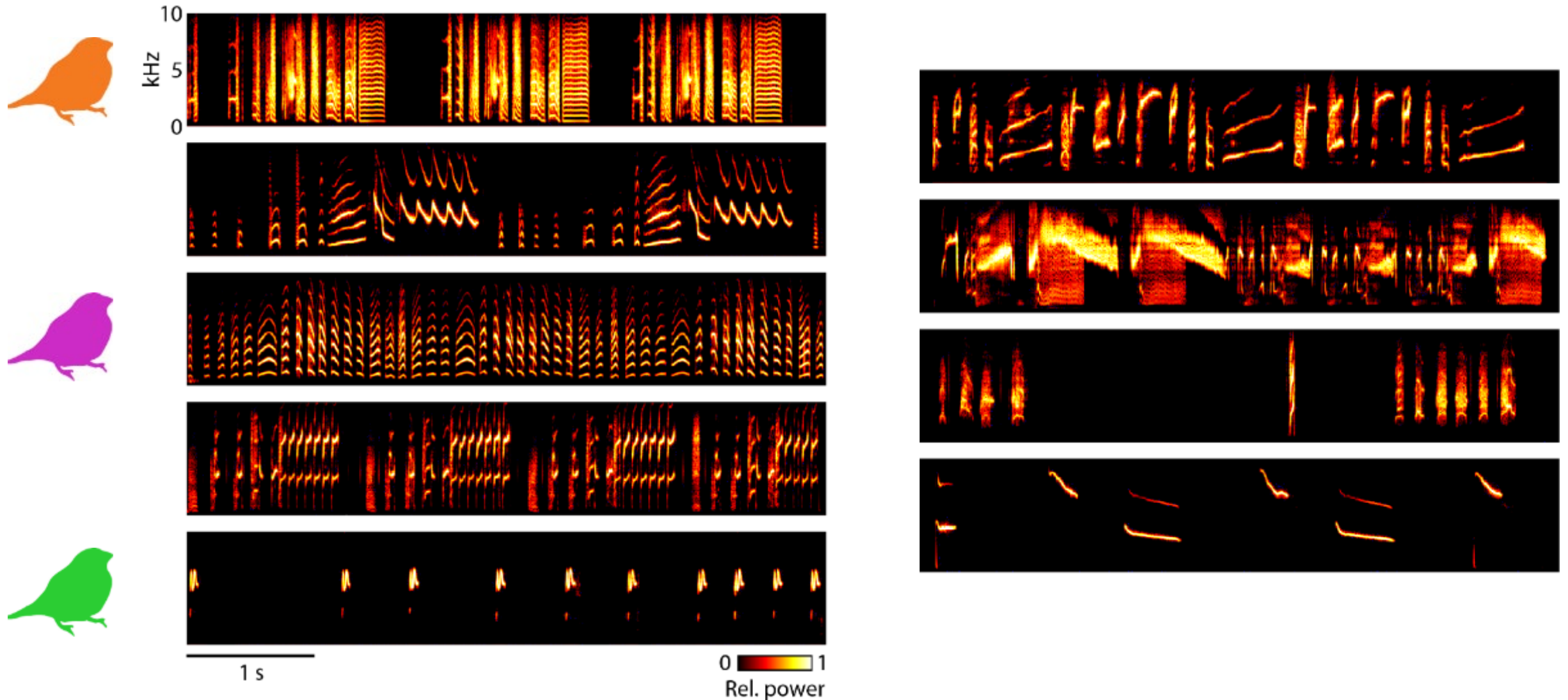
NCM is necessary for perception that supports song learning



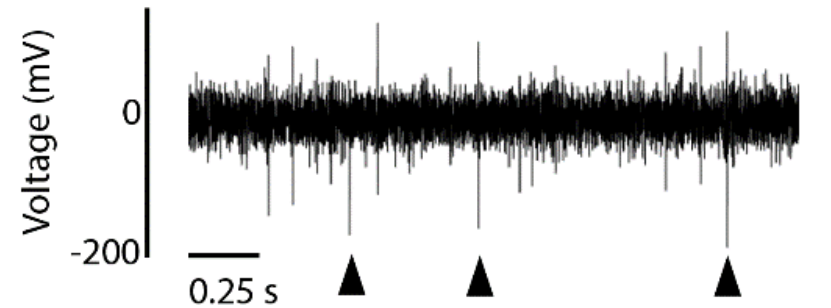
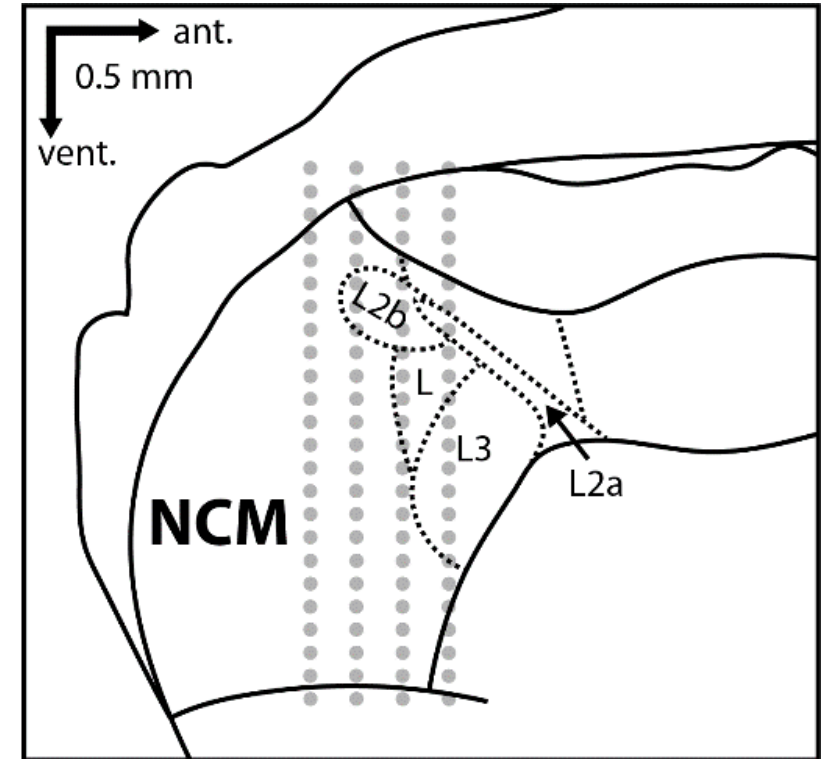
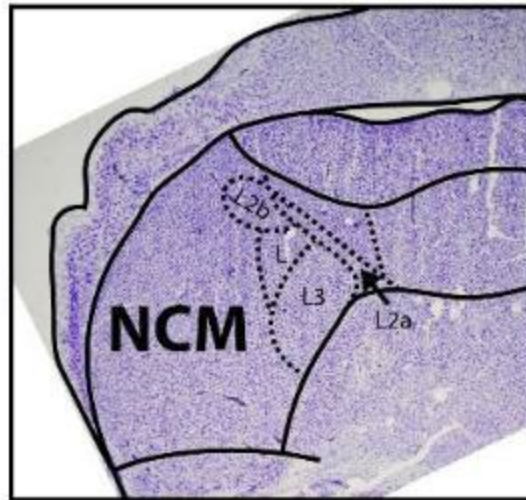
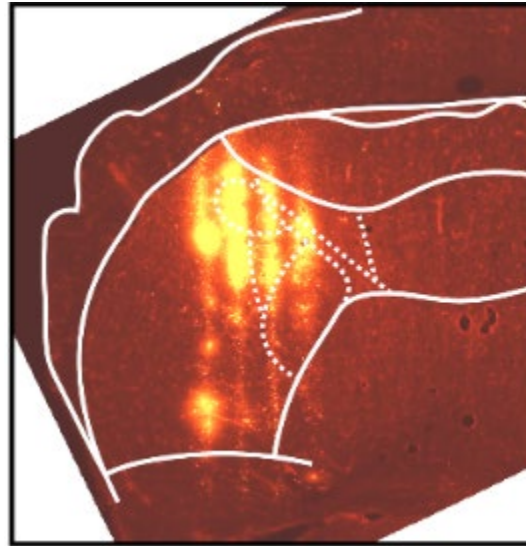
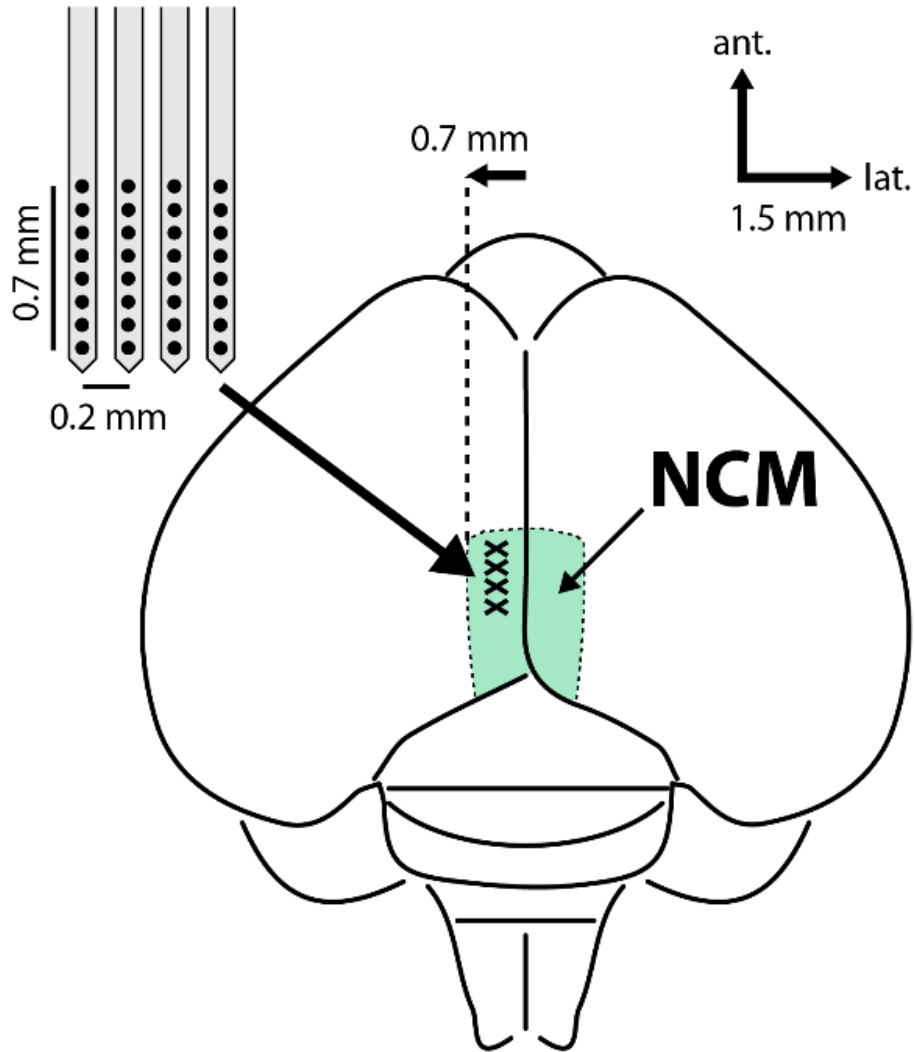
NCM neurons respond selectively to conspecific song



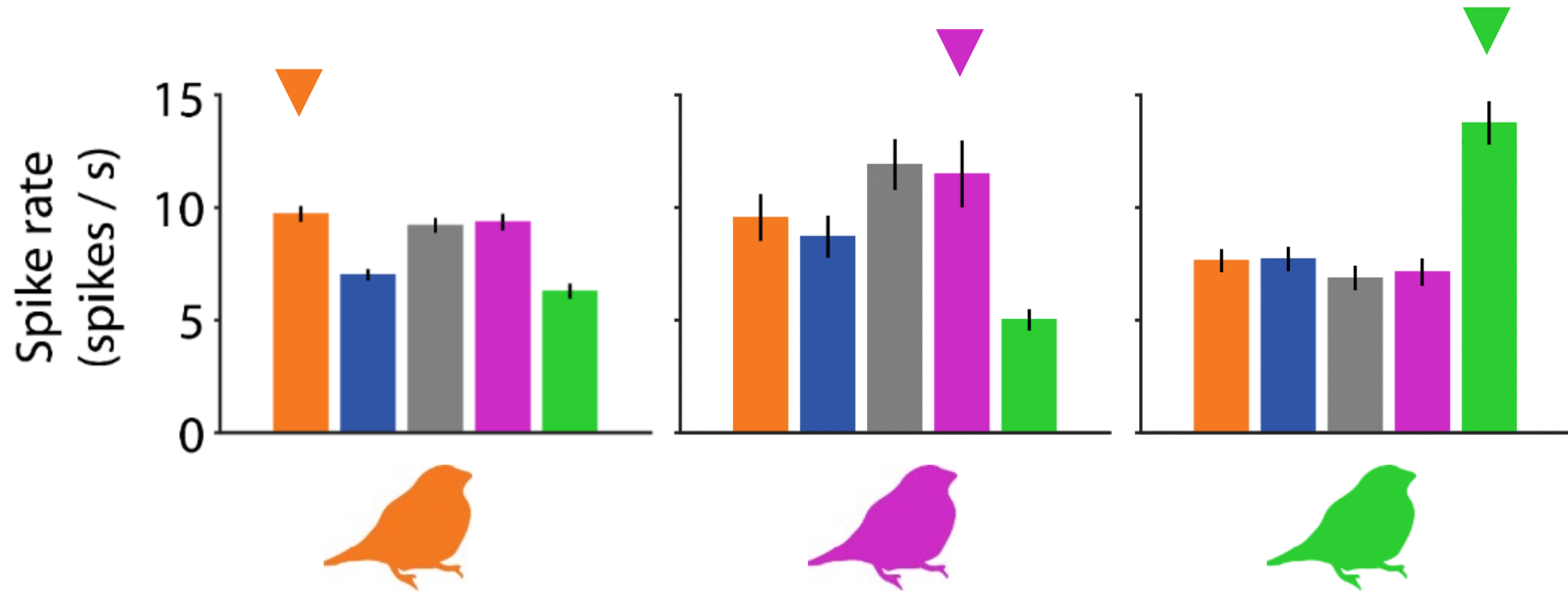
How does acoustic structure in song drive selective coding?



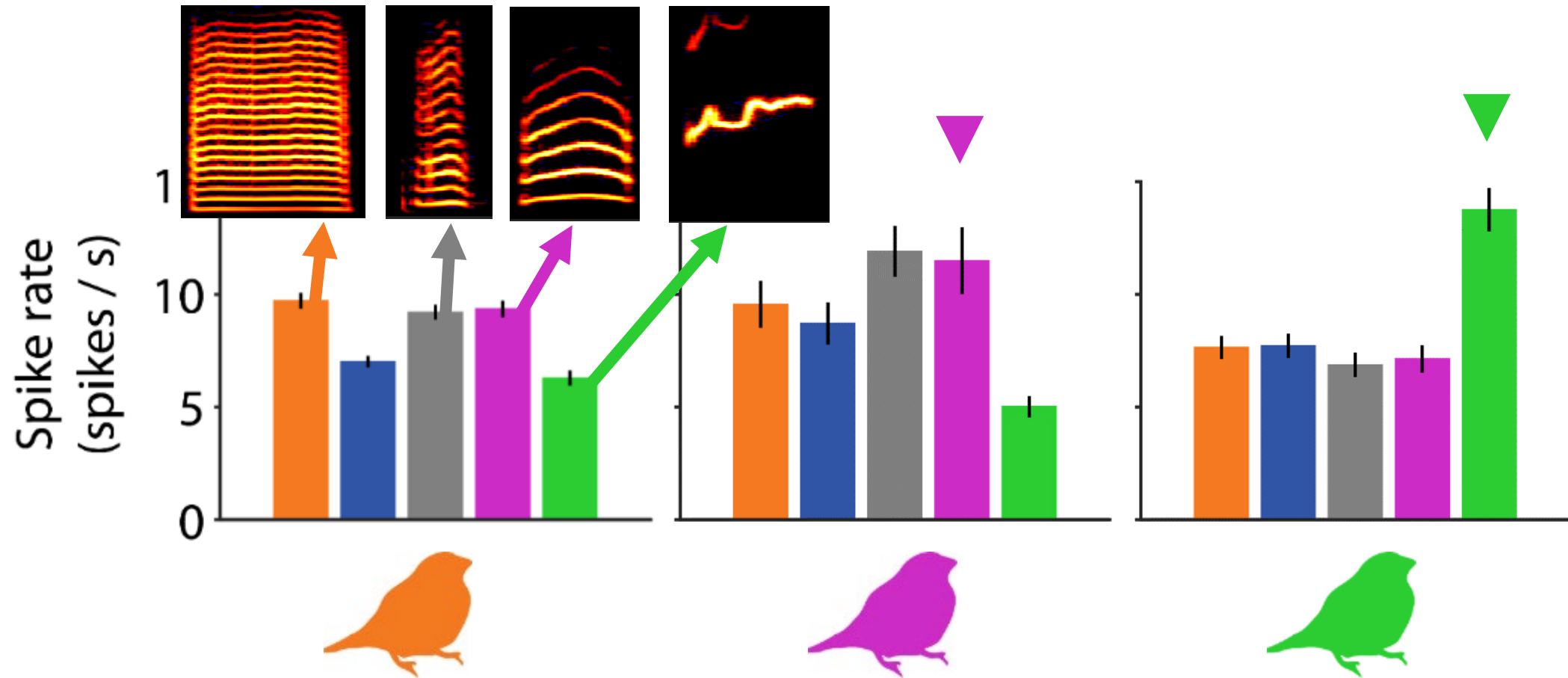
I record activity of single neurons in NCM



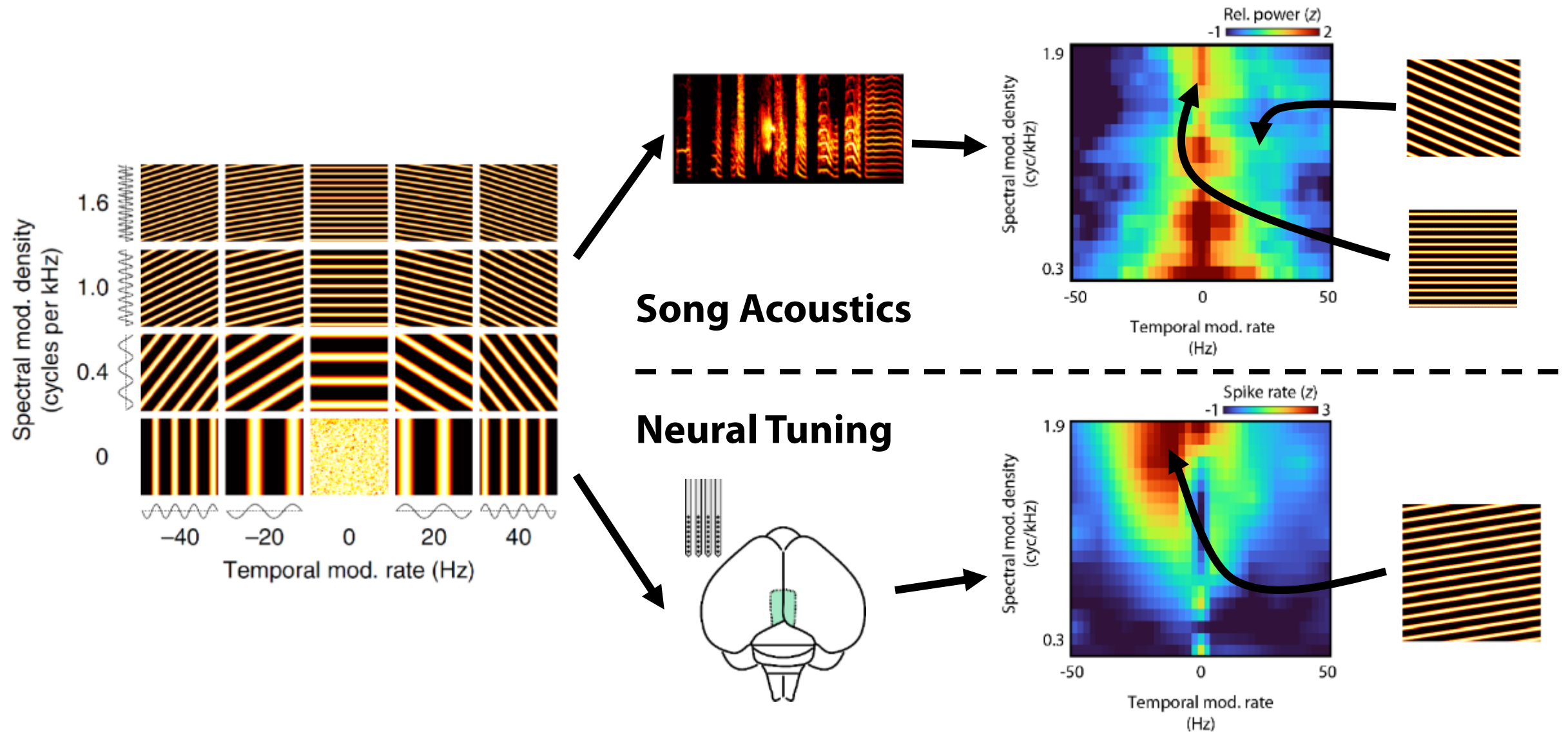
NCM neurons respond best to conspecific song... but also to some different-species songs



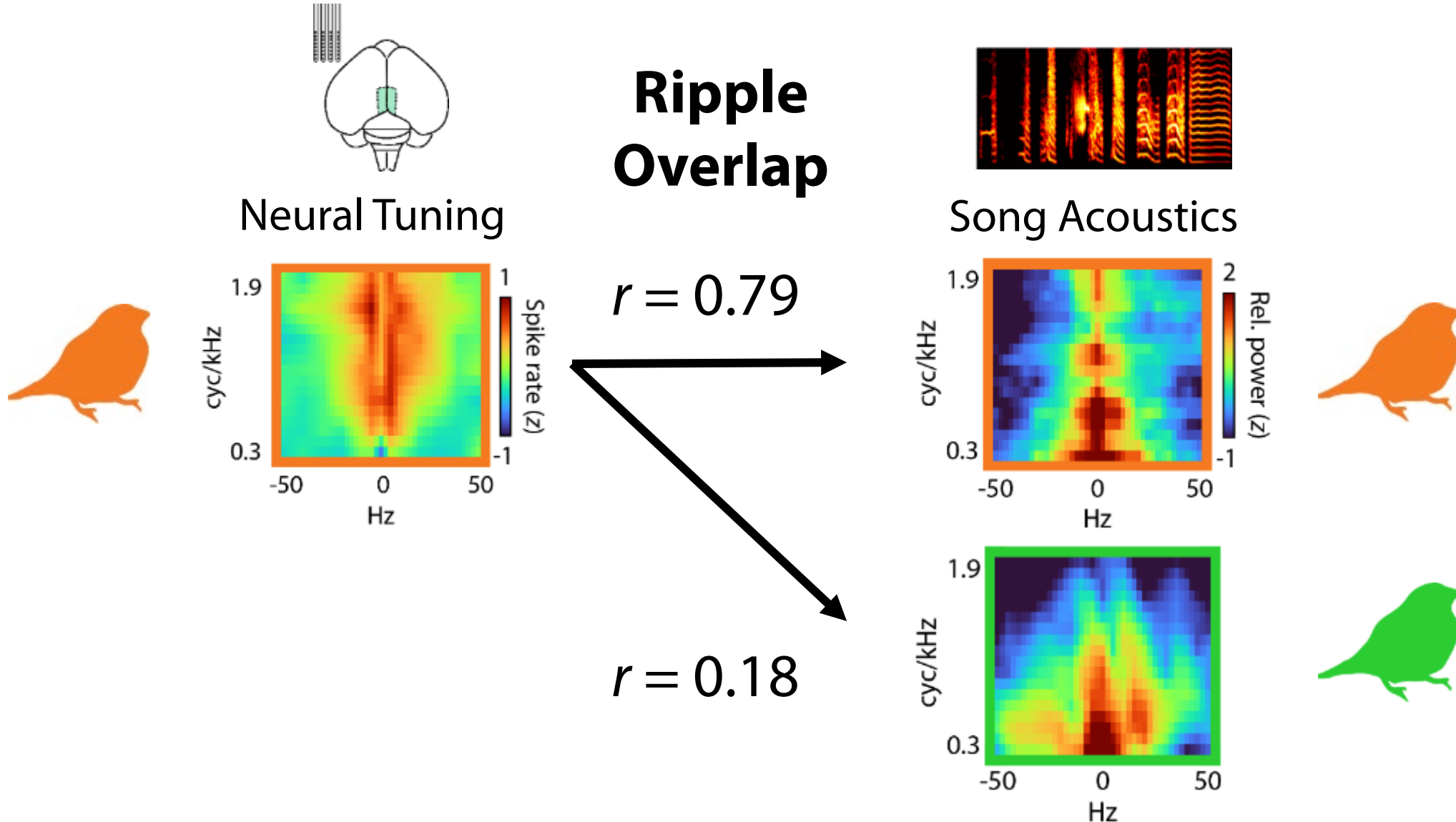
NCM neurons respond best to conspecific song... but also to some different-species songs



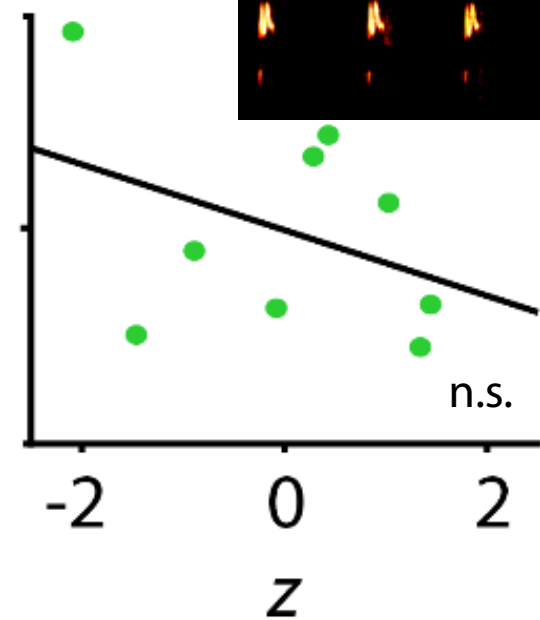
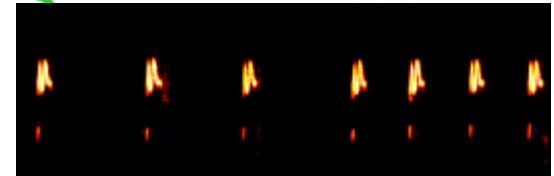
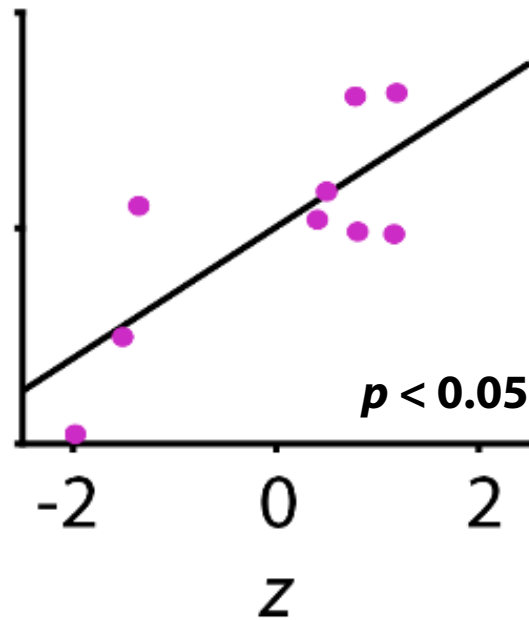
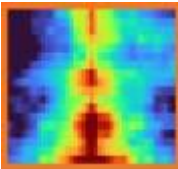
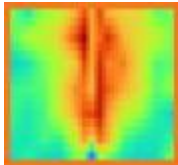
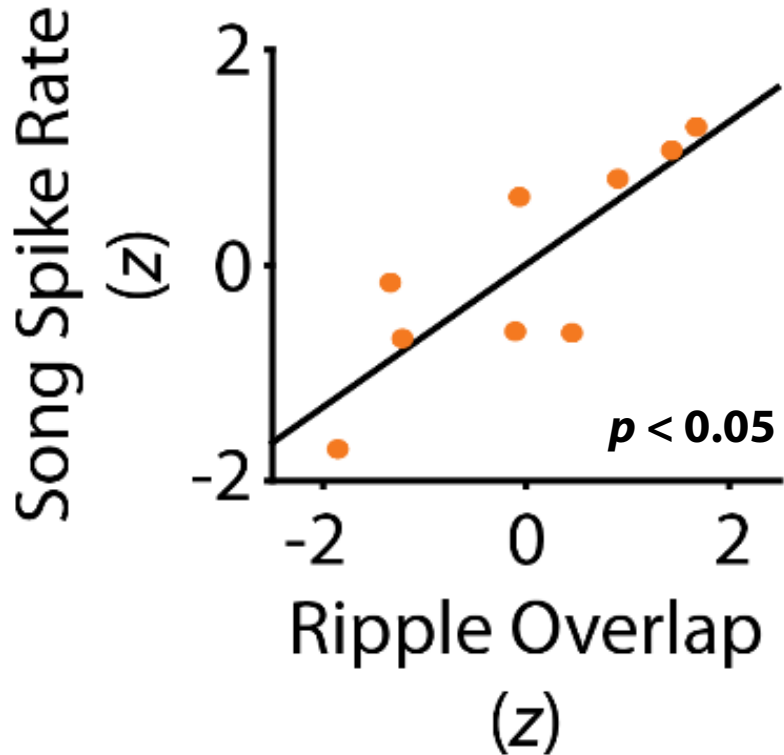
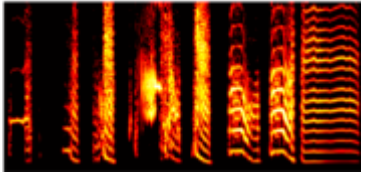
Ripple matrix links song acoustics to neural tuning



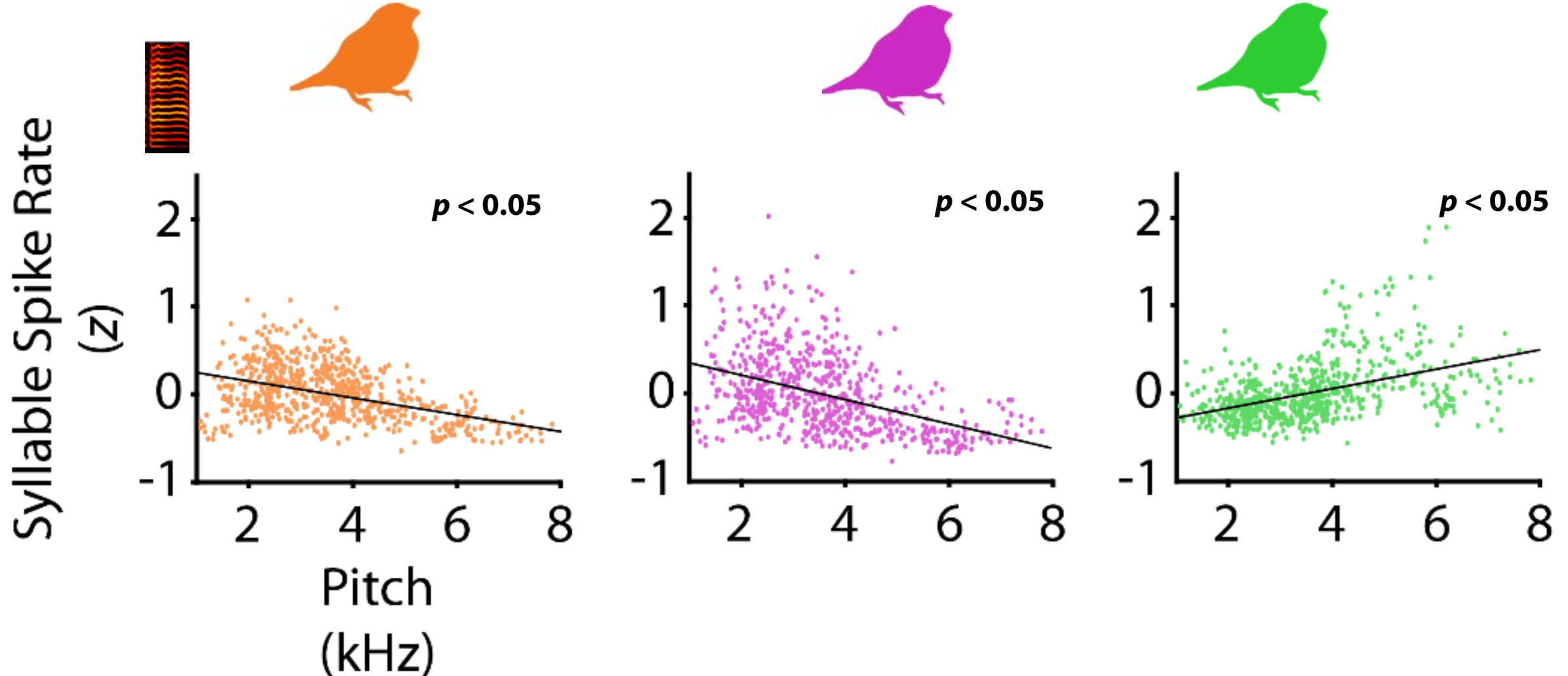
Ripple responses reveal tuning to acoustic structure



Tuning to acoustic structure explains song responses... sometimes



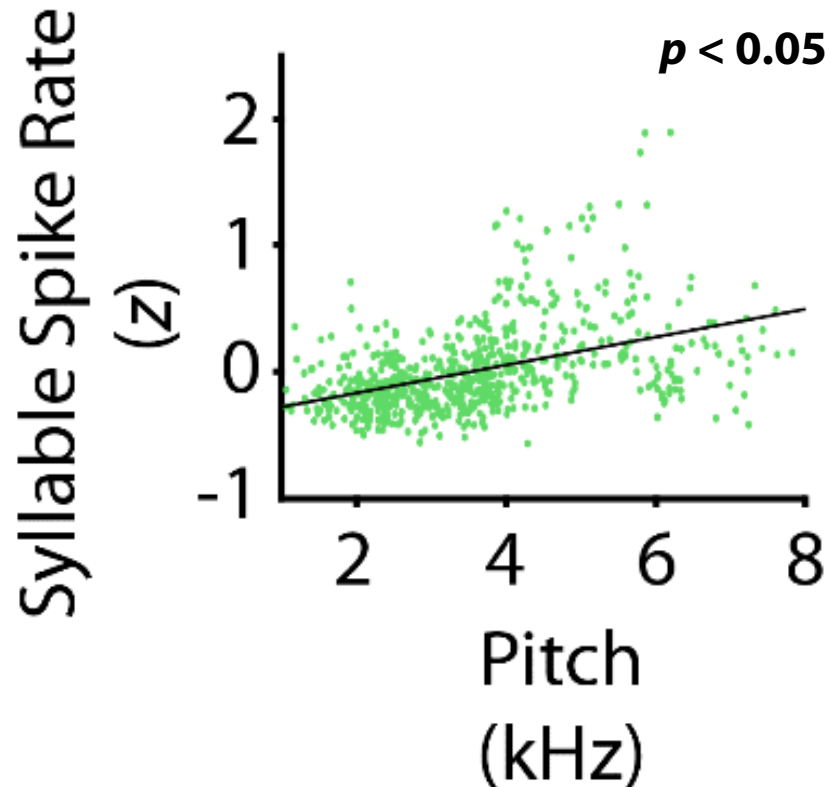
Tuning to pitch alone explains song responses in the green species



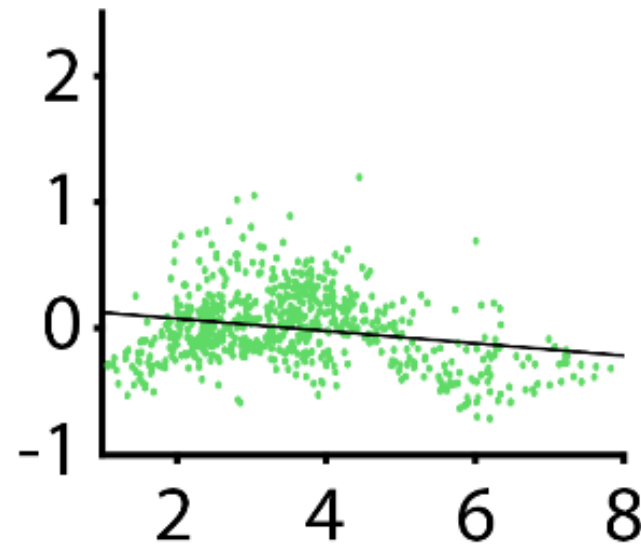
Pitch tuning in the green species is unique to **NCM**



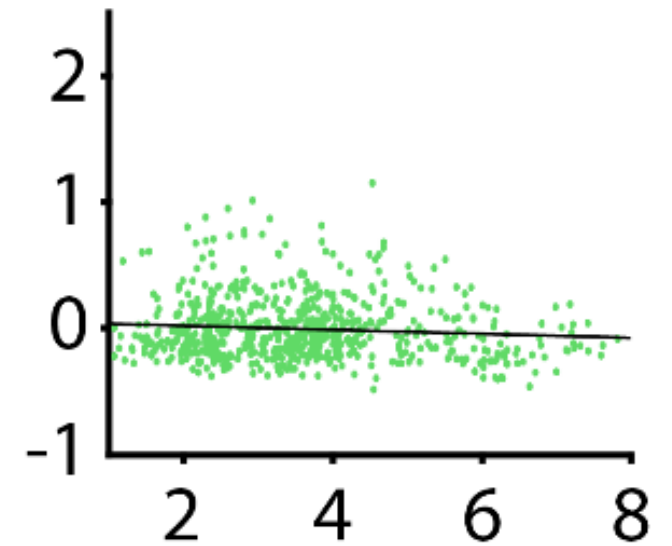
NCM



Input



NCL



Conclusions

Secondary auditory cortex neurons are tuned to the **hallmark acoustic structure** of your own species' **songs**, but they are agnostic to **species/singer identity**

Sensory tuning could align **perception** and **mating decisions** to sounds that feature **conspecific acoustic structure**



Zebra
finch



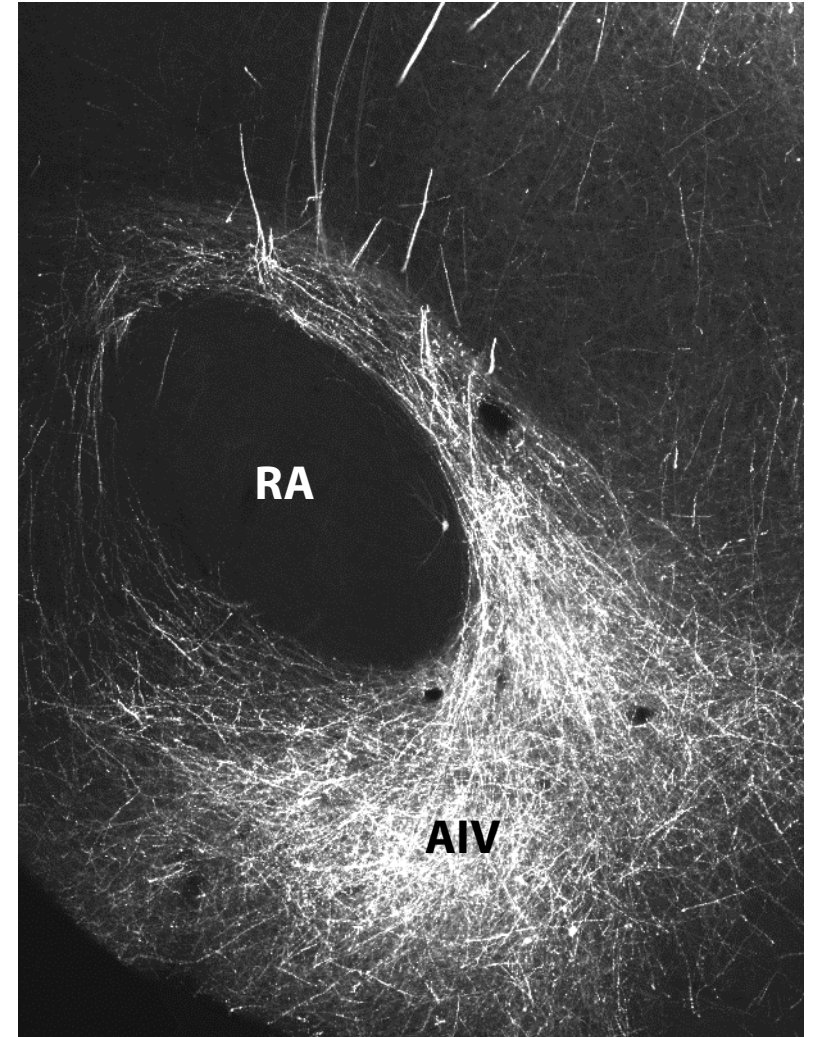
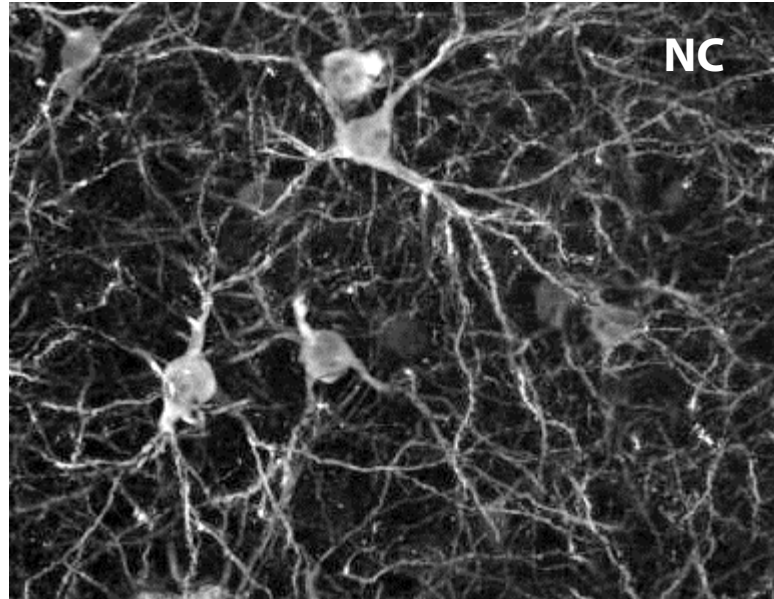
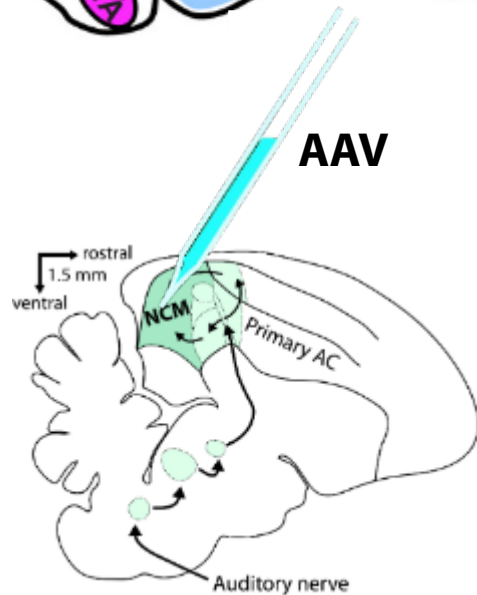
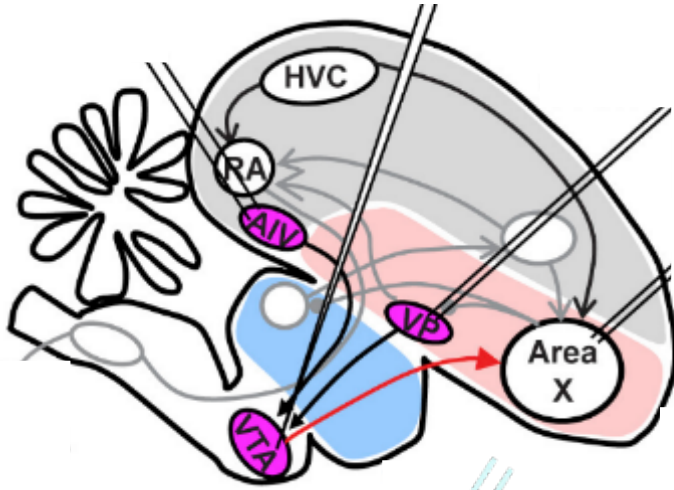
Double-barred
finch



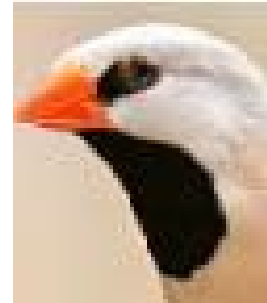
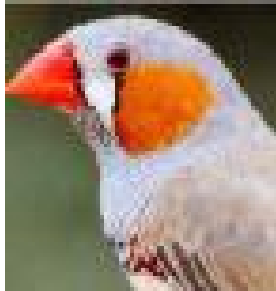
Gold-breasted
waxbill



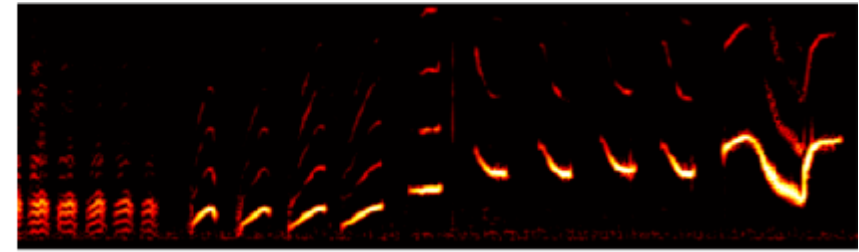
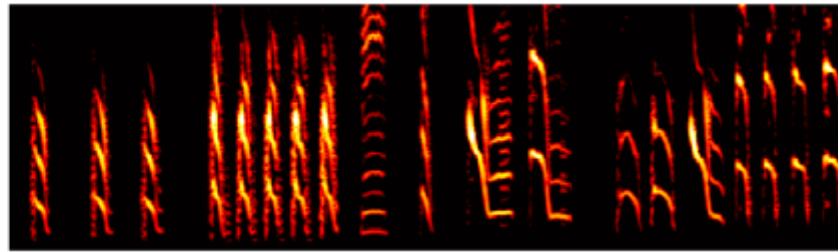
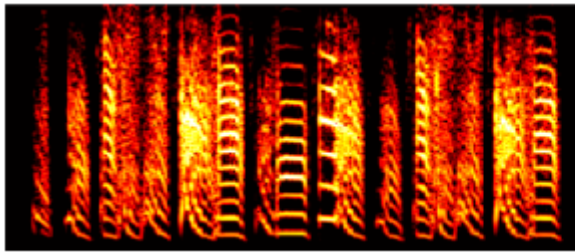
Future directions: Where does information from NCM go?



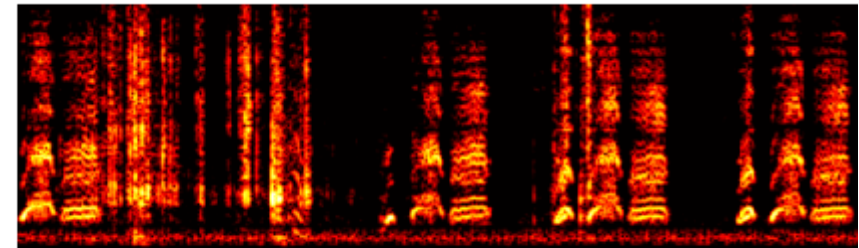
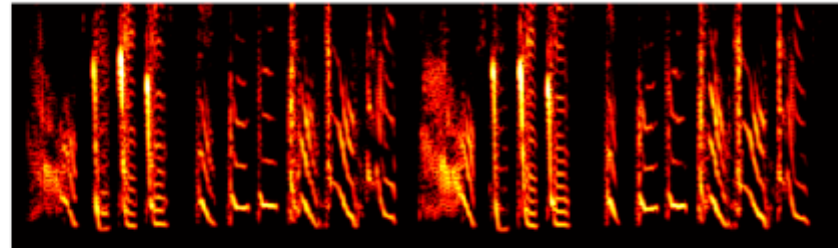
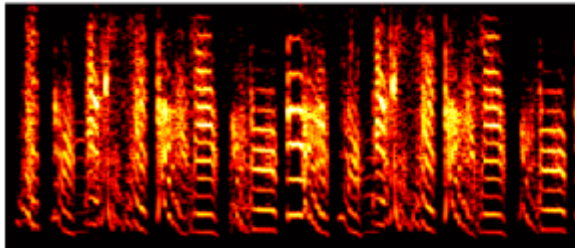
Future directions: Constraints on song learning



Tutor

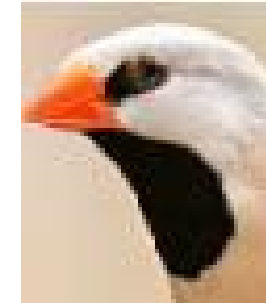
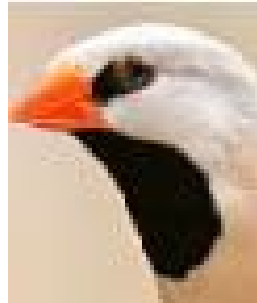
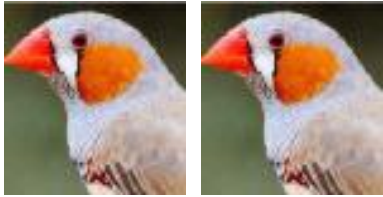


Pupil

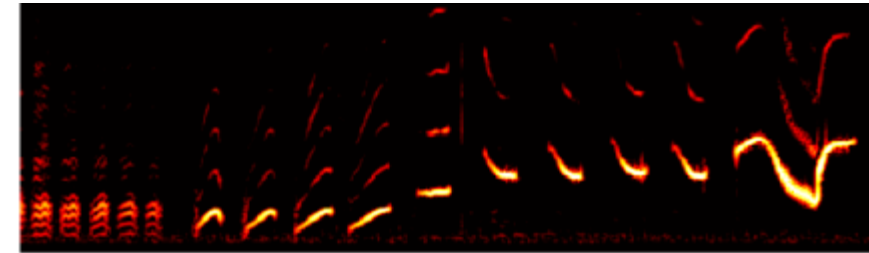
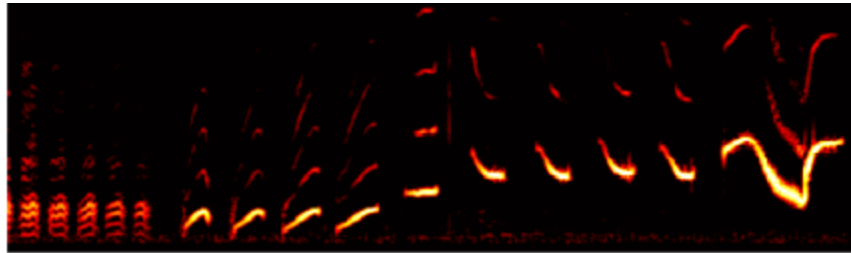


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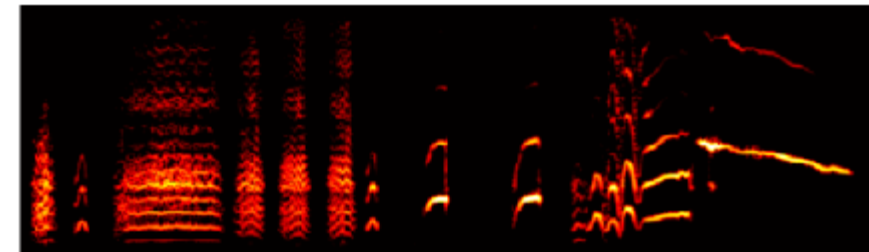
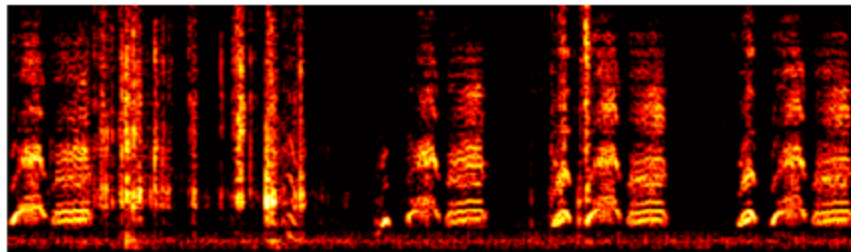
Future directions: Hybrid birds copy songs very well



Tutor



Pupil



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